PSI Center for Scientific Computing, Theory and Data

Koopmans functionals in practice

minimisation, screening coefficients, automation, and more…

Edward Linscott

ICTP–MARVEL College, 27 May 2026

Core theory

$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(\underbrace{-\int_0^{f_i} \varepsilon_i(f)\,\mathrm{d}f}_{\text{removes curvature}} + \underbrace{f_i \eta_i}_{\text{restores linearity}} \right)$$

I. Dabo *et al. Phys. Rev. B* **82,** 115121 (2010), G. Borghi *et al. Phys. Rev. B* **90,** 75135 (2014), N. Colonna *et al. J. Chem. Theory Comput.* **15,** 1905–1914 (2019)

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Differences to semi-local functionals:

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Differences to semi-local functionals:

- different flavours

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Differences to semi-local functionals:

- different flavours
- orbital-density dependence

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Differences to semi-local functionals:

- different flavours
- orbital-density dependence
- screening

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Flavours

Flavours of Koopmans functionals

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One degree of freedom: what should be the gradient of this linear term?

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- the base functional $\rightarrow$ “KI” (Koopmans integral). Enforces IP theorem. Does not affect energy/density!

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You might also see…

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- “pKIPZ” = KIPZ Hamiltonian evaluated on the KI solution

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You might also see…

- "pKIPZ" = KIPZ Hamiltonian evaluated on the KI solution
- "K" = an earlier iteration based off half-filling rather than integer endpoints (no longer used)

Orbital-density-dependence

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$$-\int_0^{f_i} \varepsilon_i(f)\,\mathrm{d}f + f_i \int_0^1 \varepsilon_i(f)\,\mathrm{d}f = E_{\mathrm{Hxc}}[\rho] + E_{\mathrm{Hxc}}[\rho - \rho_i] + f_i(-E_{\mathrm{Hxc}}[\rho - \rho_i] + E_{\mathrm{Hxc}}[\rho - \rho_i + n_i])$$

Orbital-density-dependence

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For filled orbitals with KI:

$$v_i^{\mathrm{KI}}/\alpha_i = -E_{\mathrm{H}}[n_i] + E_{\mathrm{xc}}[\rho] - E_{\mathrm{xc}}[\rho-n_i] - \int \mathrm{d}\boldsymbol{r}'\, v_{\mathrm{xc}}(\boldsymbol{r}',[\rho])\, n_i(\boldsymbol{r}')$$

Orbital-density-dependence

Initialise with MLWFs, then (optionally) solve with CG minimisation:

G. Borghi *et al. Phys. Rev. B* **91,** 155112 (2015)

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Gives rise to a set of **minimising orbitals** (localised/variational); diagonalising at the minimum gives rise to **diagonalising orbitals** (delocalised/canonical)

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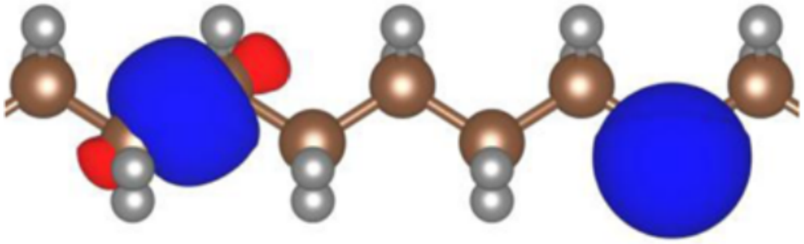
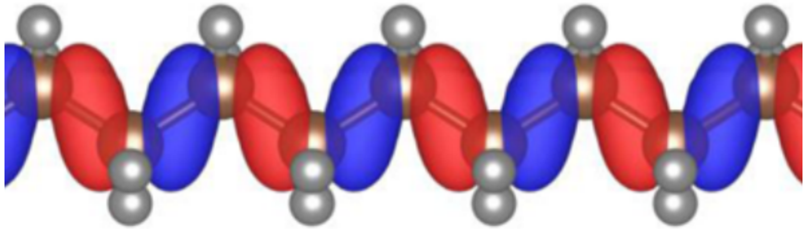
Orbital-density-dependence



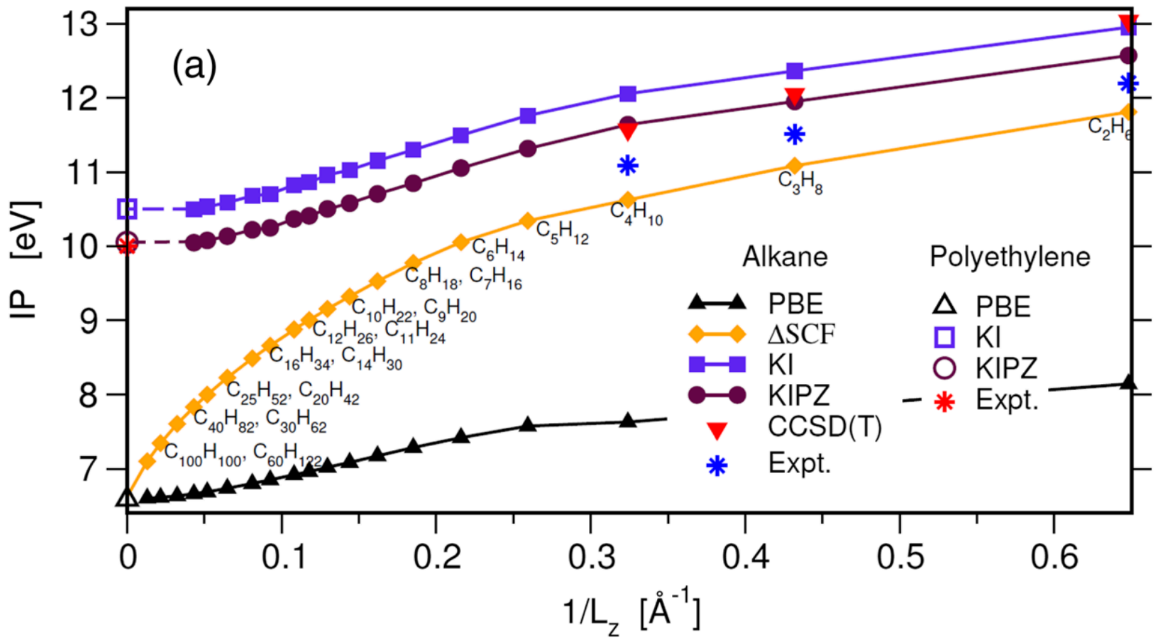
variational



canonical

N. L. Nguyen *et al. Phys. Rev. X* **8,** 21051 (2018)

Importance of localisation



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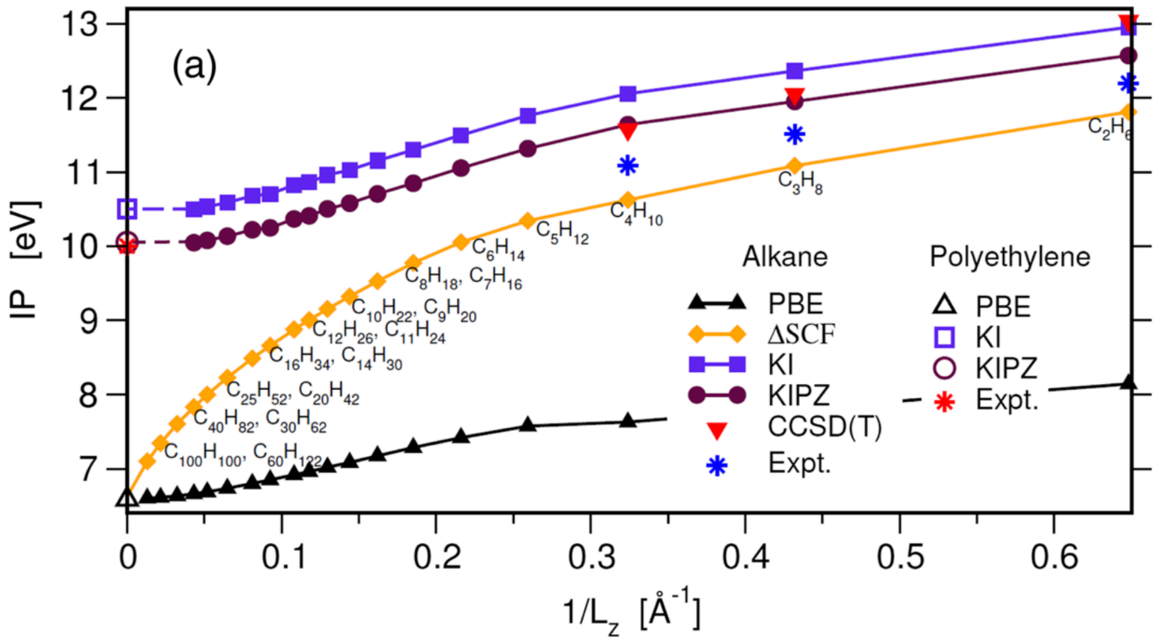

In the bulk limit for one cell $\Delta E = E(N - \delta N) - E(N)$

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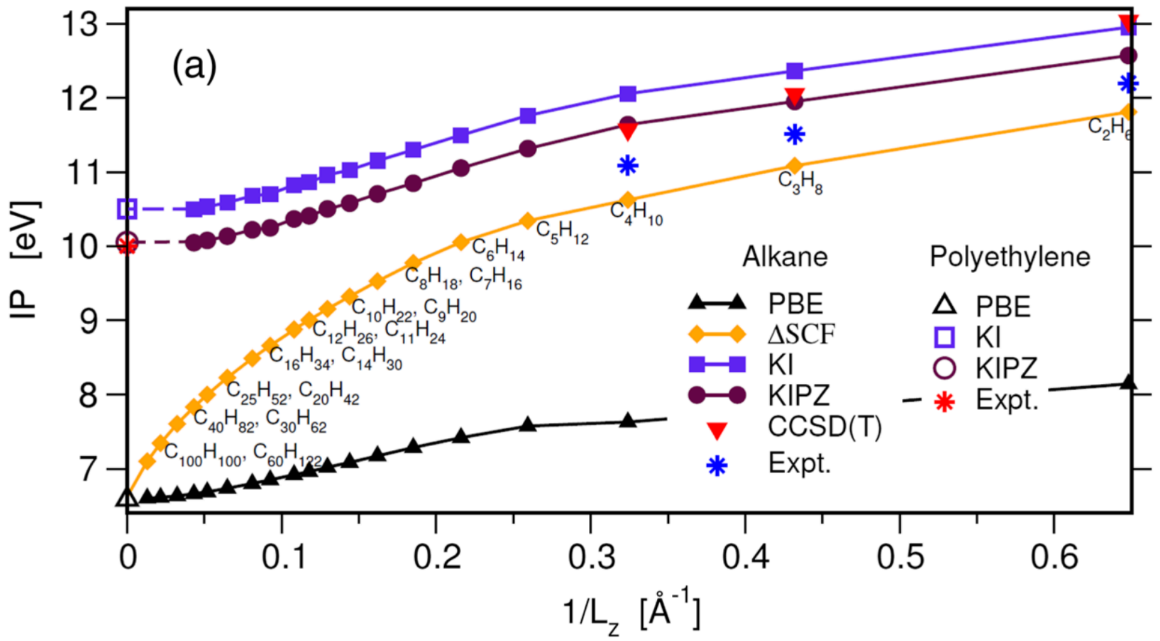

In the bulk limit for one cell $\Delta E = E(N - \delta N) - E(N)$

Across all the cells $\Delta E = \frac{1}{\delta N}(E(N - \delta N) - E(N)) = -\frac{\mathrm{d}E}{\mathrm{d}N} = -\varepsilon_{\mathrm{HO}}$

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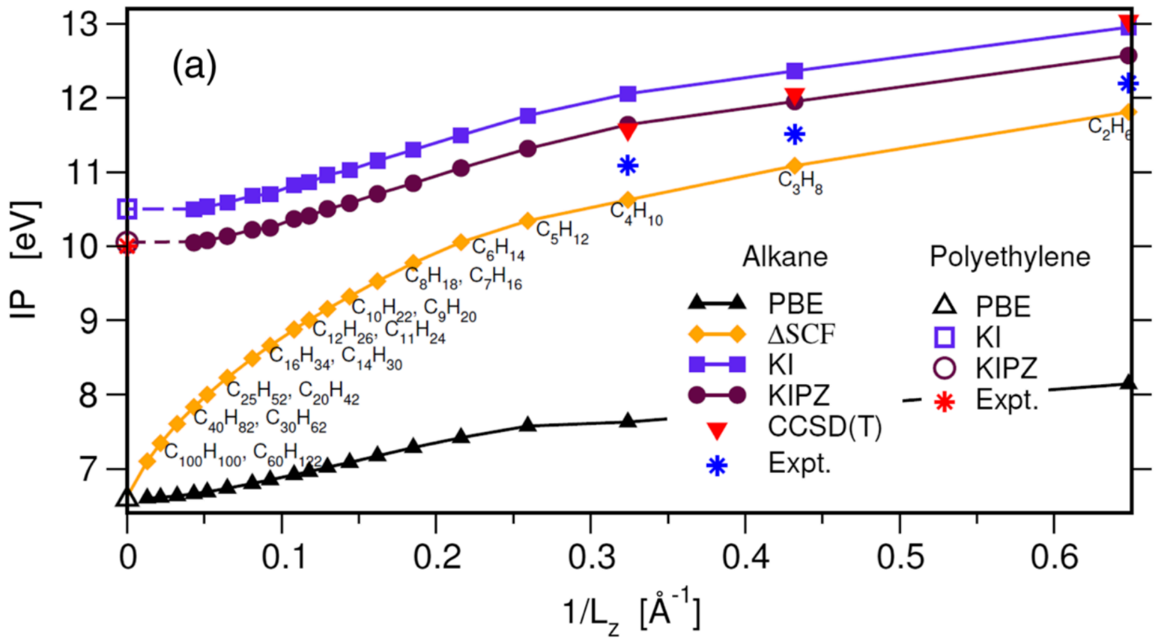

$$\lim_{n_i(r)\to 0} v_i^{\mathrm{KI}}/\alpha_i = \lim_{n_i(r)\to 0} -E_{\mathrm{H}}[n_i] + E_{\mathrm{xc}}[\rho] - E_{\mathrm{xc}}[\rho - n_i] - \int \mathrm{d}\boldsymbol{r}'\, v_{\mathrm{xc}}(\boldsymbol{r}', [\rho])\, n_i(\boldsymbol{r}') = 0$$

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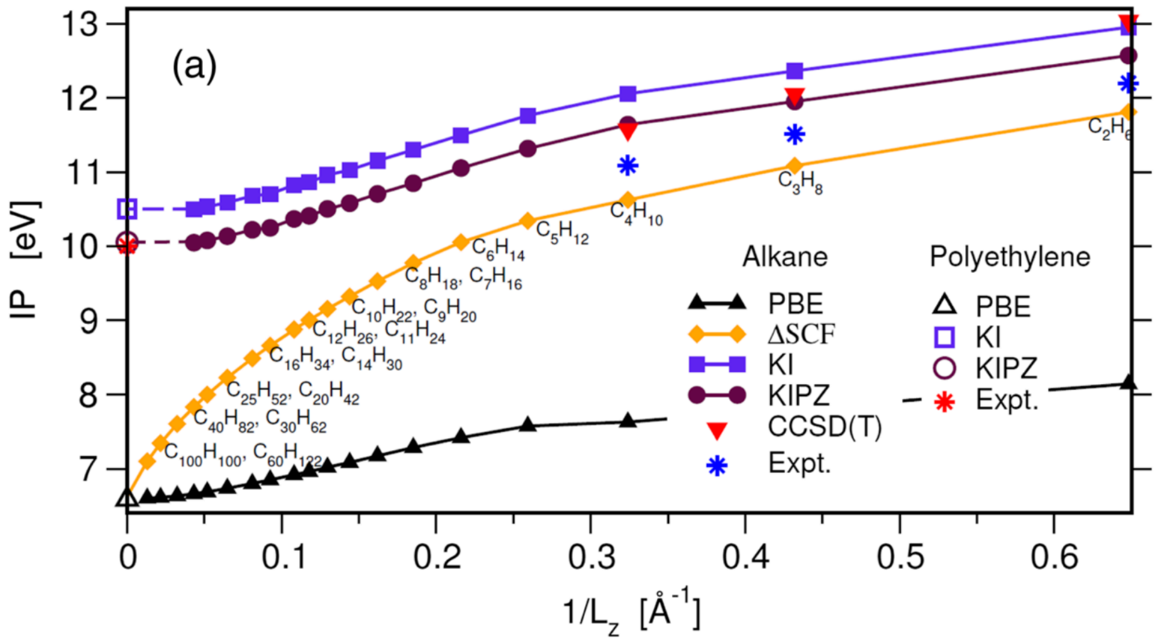

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Solution: apply Koopmans correction to localised orbitals

N. L. Nguyen *et al. Phys. Rev. X* **8,** 21051 (2018)

Orbital-density-dependence

Other features of orbital-density-dependence

A. Ferretti *et al. Phys. Rev. B* **89,** 195134 (2014)

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Other features of orbital-density-dependence

- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general

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- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general
- Practically we can often use MLWFs
- a natural generalisation in the direction of spectral functional theory

A. Ferretti *et al. Phys. Rev. B* **89,** 195134 (2014)

Screening

Screening

- In Hartree-Fock (the original “Koopmans’ theorem”)

$$E_{\mathrm{ee}}^{\mathrm{HF}} = \frac{1}{2}\sum_{ij} f_i f_j \int \mathrm{d}\boldsymbol{r}\,\mathrm{d}\boldsymbol{r}' \frac{|\psi_i(\boldsymbol{r})|^2\ |\psi_j(\boldsymbol{r}')|^2}{\boldsymbol{r}-\boldsymbol{r}'} - \frac{\psi_i^*(\boldsymbol{r})\psi_j^*(\boldsymbol{r}')\psi_i(\boldsymbol{r}')\psi_j(\boldsymbol{r})}{\boldsymbol{r}-\boldsymbol{r}'}$$

C. Li *et al. J. Chem. Phys.* **146,** 74107 (2017)

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$$\frac{d^2 E_{\mathrm{ee}}^{\mathrm{HF}}}{df_i df_j} \stackrel{?}{=} 0$$

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PSI

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$$= E_{\mathrm{DFT}}[\rho] + \sum_i \left(-\left(E_{\mathrm{DFT}}[\rho] - E_{\mathrm{DFT}}\left[\rho^{f_i\to 0}\right]\right) + f_i\left(E_{\mathrm{DFT}}\left[\rho^{f_i\to 1}\right] - E_{\mathrm{DFT}}\left[\rho^{f_i\to 0}\right]\right)\right)$$

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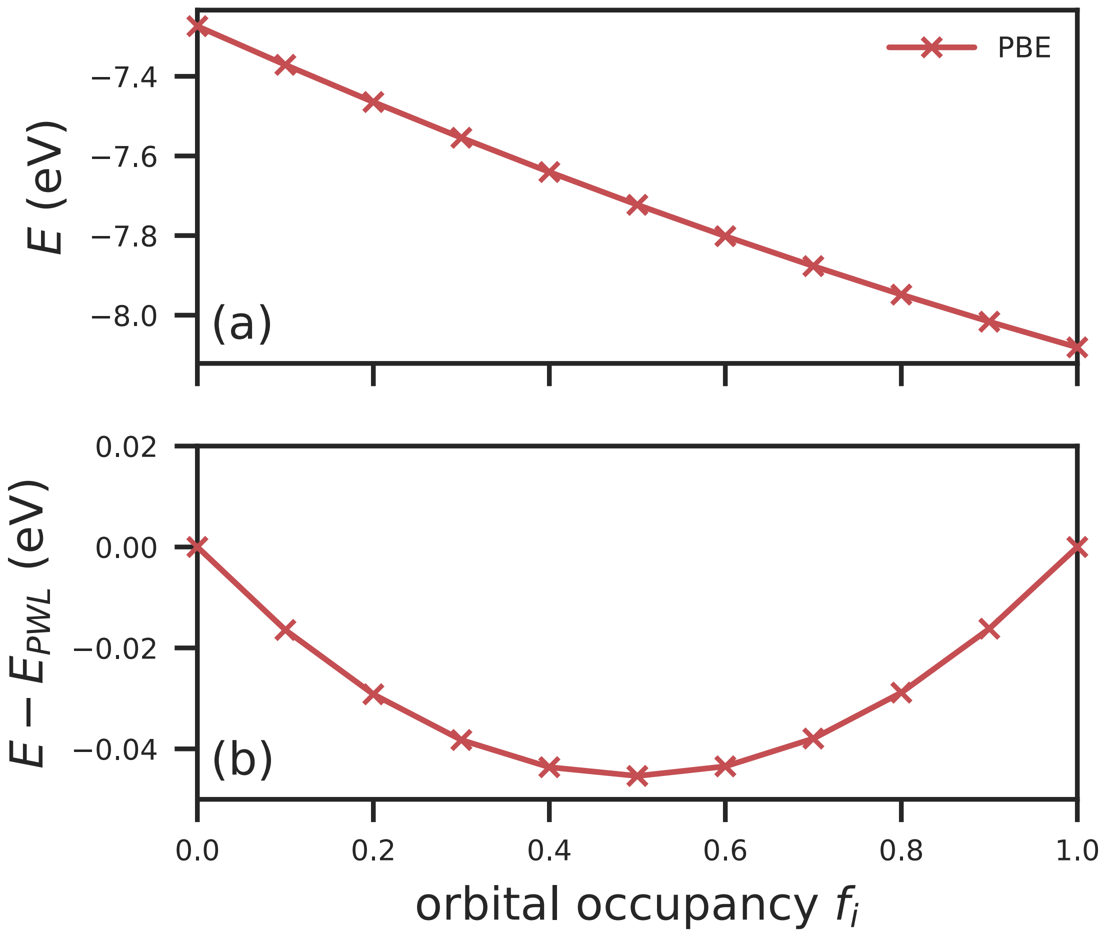
total energy differences

$$= E_{\mathrm{DFT}}[\rho] + \sum_i \left(-\left(E_{\mathrm{DFT}}[\rho] - E_{\mathrm{DFT}}\left[\rho^{f_i\to 0}\right]\right) + f_i\left(E_{\mathrm{DFT}}\left[\rho^{f_i\to 1}\right] - E_{\mathrm{DFT}}\left[\rho^{f_i\to 0}\right]\right)\right)$$

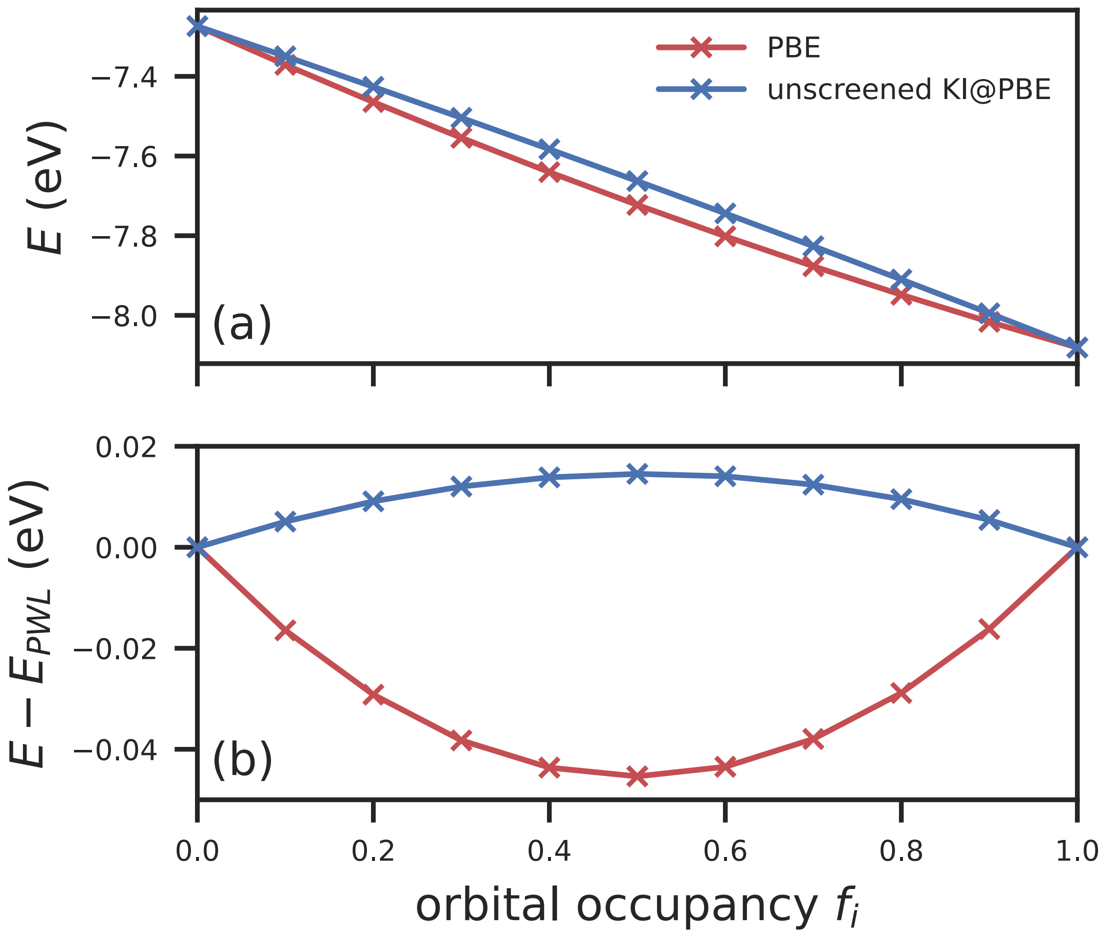
cannot be evaluated directly

C. Li *et al.* *J. Chem. Phys.* **146,** 74107 (2017)

Screening


PBE
(a)
(b)
E (eV)
−7.4
−7.6
−7.8
−8.0
E − E_PWL (eV)
0.02
0.00
−0.02
−0.04
0.0
0.2
0.4
0.6
0.8
1.0
orbital occupancy f_i

Screening



PBE
unscreened KI@PBE
E (eV)
−7.4
−7.6
−7.8
−8.0
(a)
$E - E_{PWL}$ (eV)
0.02
0.00
−0.02
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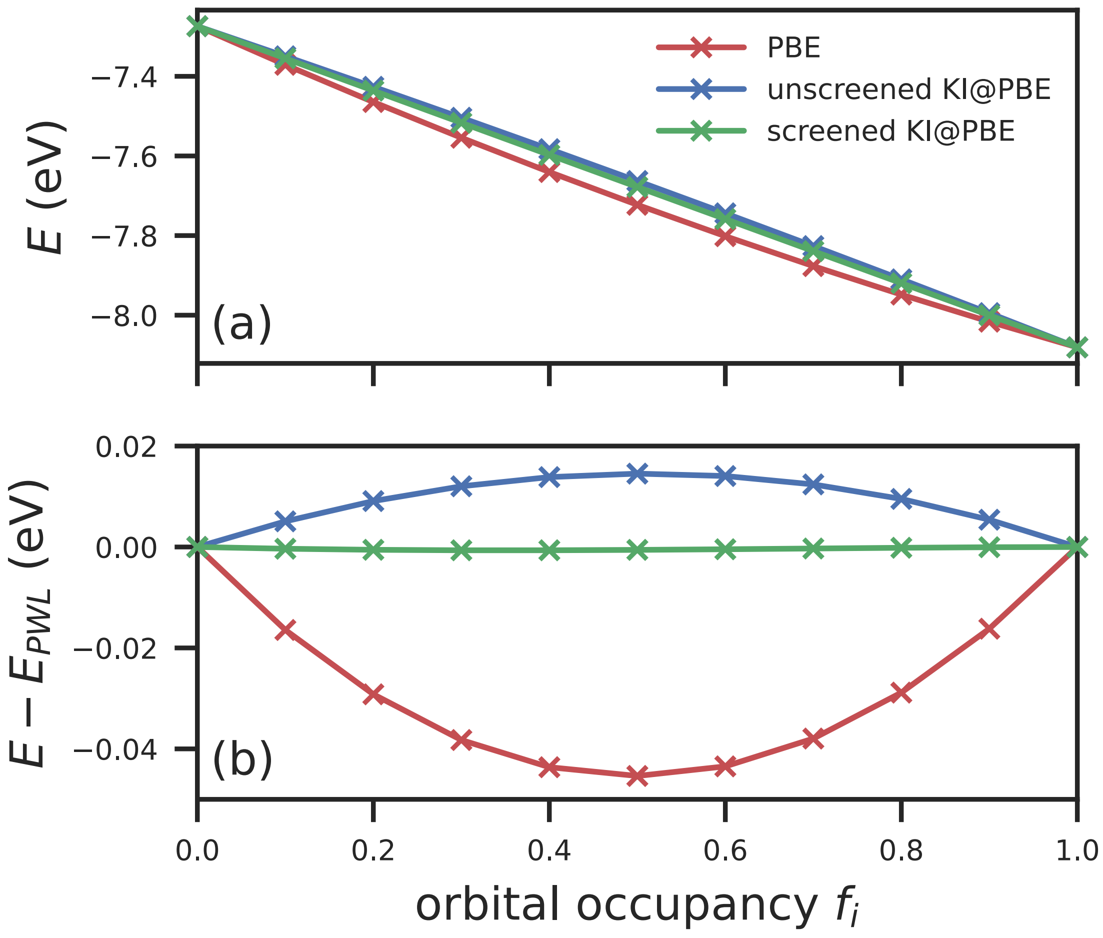
PBE
unscreened KI@PBE
screened KI@PBE
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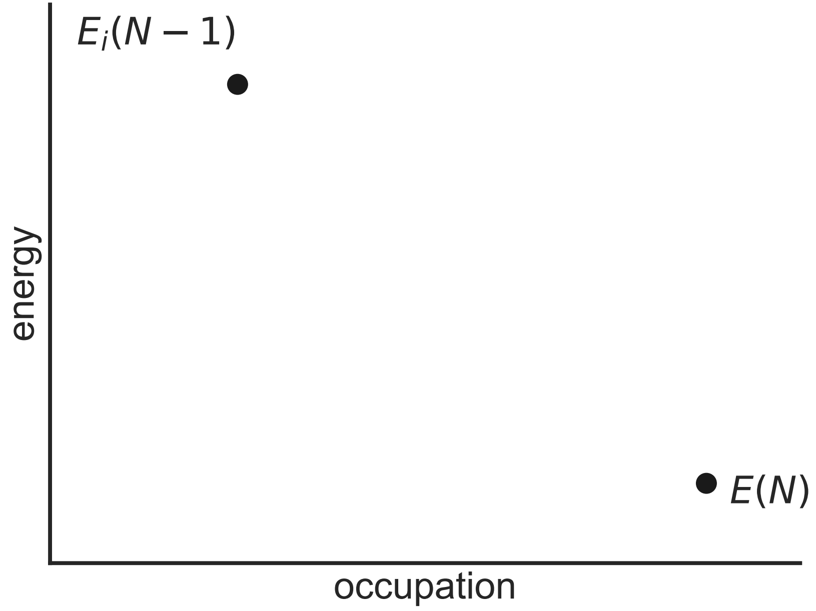

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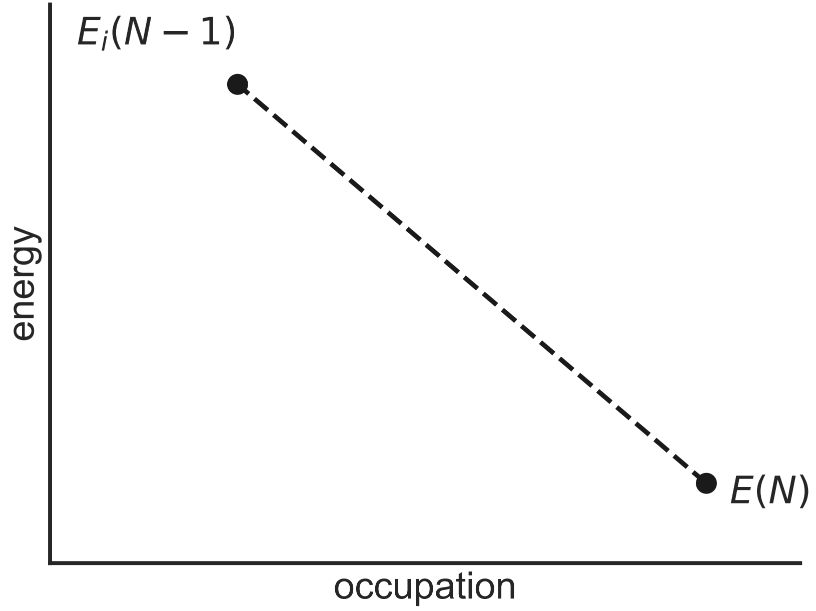

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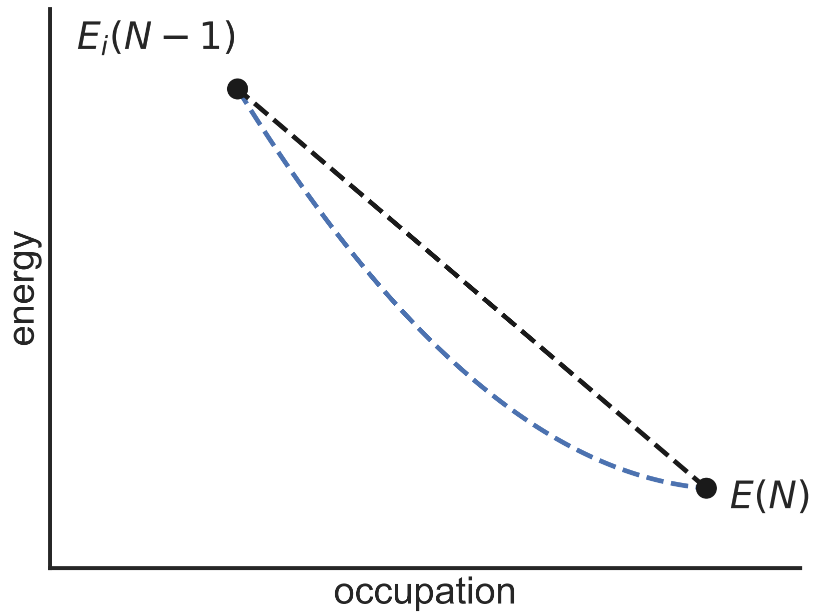
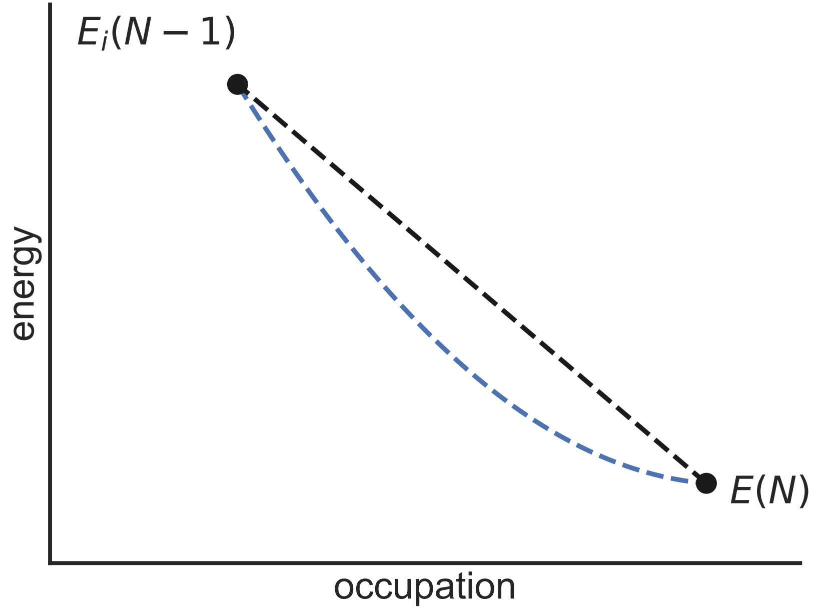


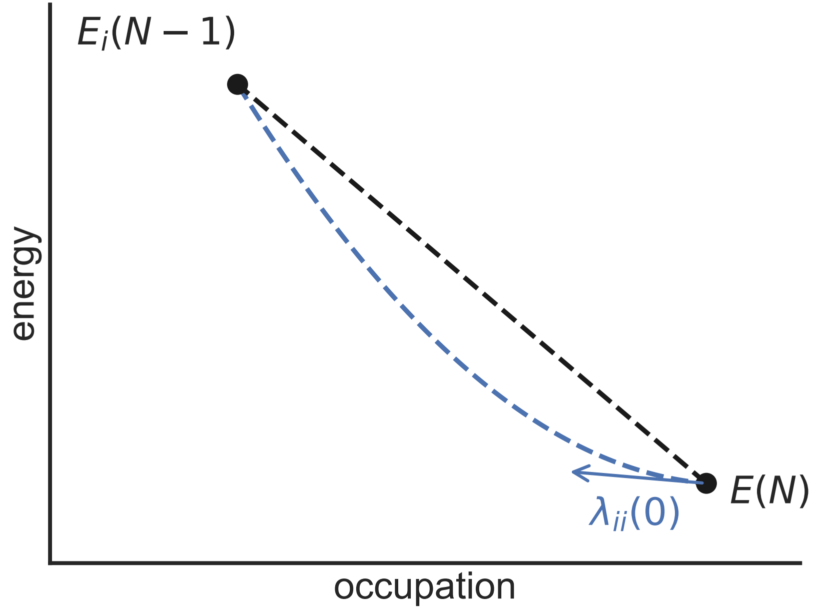
Screening

PSI



$$\lambda_{ii}(\alpha) \equiv \left\langle \varphi_i \mid \hat{h}^{\mathrm{DFT}} + \alpha \hat{v}^{\mathrm{Koopmans}} \mid \varphi_i \right\rangle = \left.\frac{\mathrm{d}E^{\mathrm{Koopmans}}}{\mathrm{d}f_i}\right|_{f_i=s}$$

Screening



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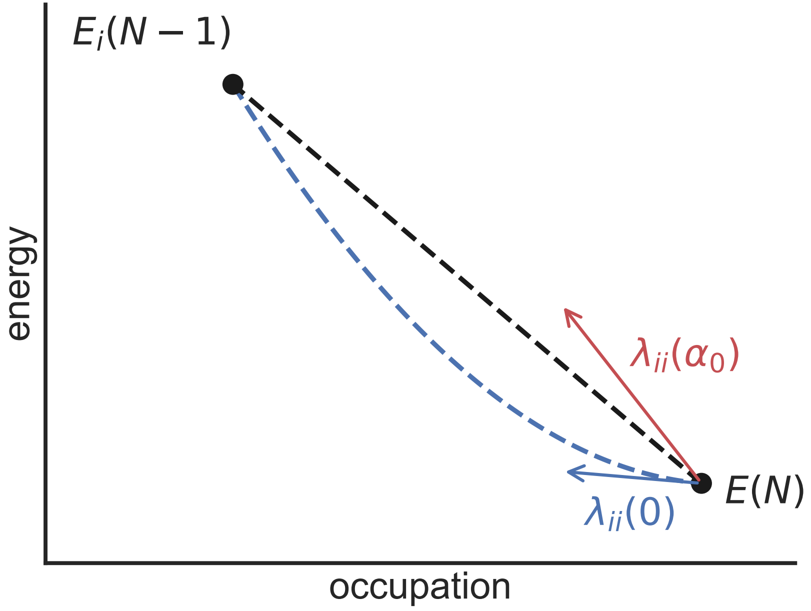

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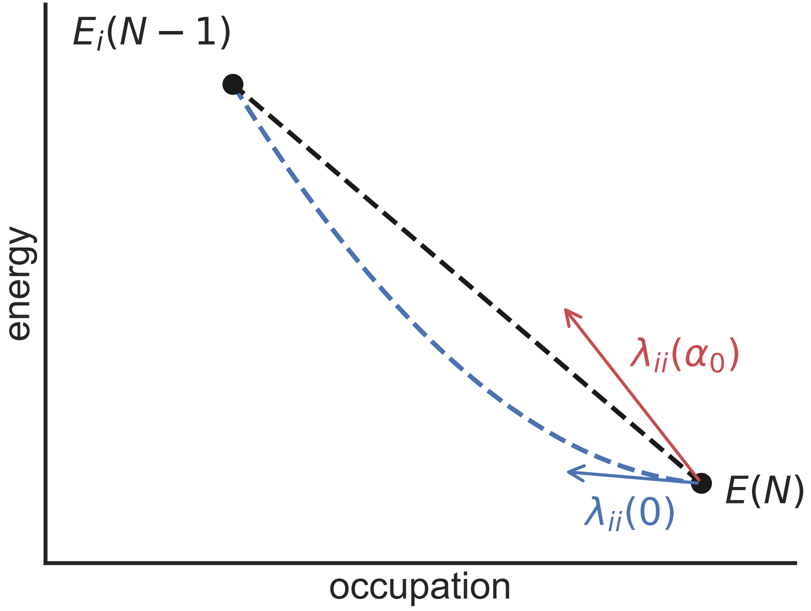
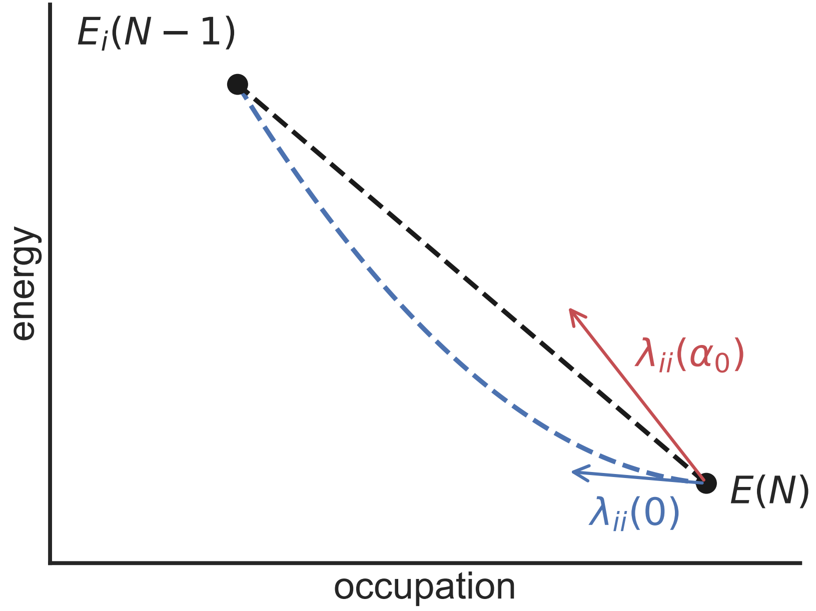


Given this, how to work out the ideal α?

Screening



Given this, how to work out the ideal α?

$$\alpha_{n+1} = \alpha_n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha_n) - \lambda_{ii}(0)}$$

Screening

PSI

$$\alpha^{n+1} = \underset{\uparrow\atop \text{screening parameter}}{\alpha^n} \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

Screening

PSI

total energy with electron
removed from orbital i

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

screening parameter

Screening

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expectation value of $\hat{H}^{\text{DFT}}$

Screening via DFPT

How can we avoid explicit charged defect calculations in a supercell?

N. Colonna *et al. J. Chem. Theory Comput.* **15,** 1905–1914 (2019)

N. Colonna *et al. J. Chem. Theory Comput.* **18,** 5435–5448 (2022)

R. De Gennaro *et al. Phys. Rev. B* **106,** 35106 (2022)

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Reformulate in terms of DFPT…

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… in reciprocal space

$$\alpha_{\mathbf{0}i} = 1 + \frac{\sum_{\boldsymbol{q}} \langle v^{\mathbf{0}i}_{\text{pert},\boldsymbol{q}} \mid \Delta^{\mathbf{0}i}_{\boldsymbol{q}} n\rangle}{\sum_{\boldsymbol{q}} \langle n^{\mathbf{0}i}_{\boldsymbol{q}} \mid v^{\mathbf{0}i}_{\text{pert},\boldsymbol{q}}\rangle}$$

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N.B. even for the supercell, we can still reconstruct a band structure

N. Colonna *et al.* *J. Chem. Theory Comput.* **15,** 1905–1914 (2019)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18,** 5435–5448 (2022)

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  - are we going to explicitly minimise the ODD?

To summarise…

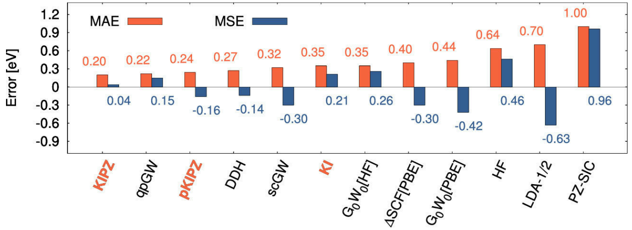
When performing a Koopmans calculation, you must decide…

- which flavour (KI, KIPZ)?
- how are we treating the orbital-density-dependence?
  - how are we initialising our variational orbitals? (N.B. depends on the flavour!)
  - are we going to explicitly minimise the ODD?
- how are we calculating the screening parameters? (finite differences, DFPT, ML…)

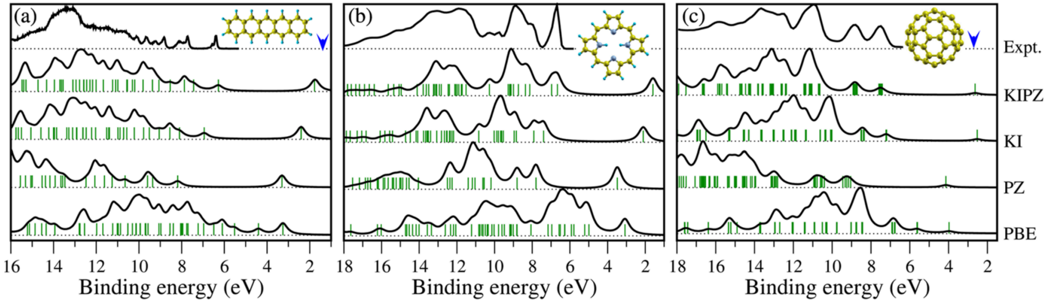
Results

Molecules

Ionisation potentials of 100 molecules cf. CCSD(T)

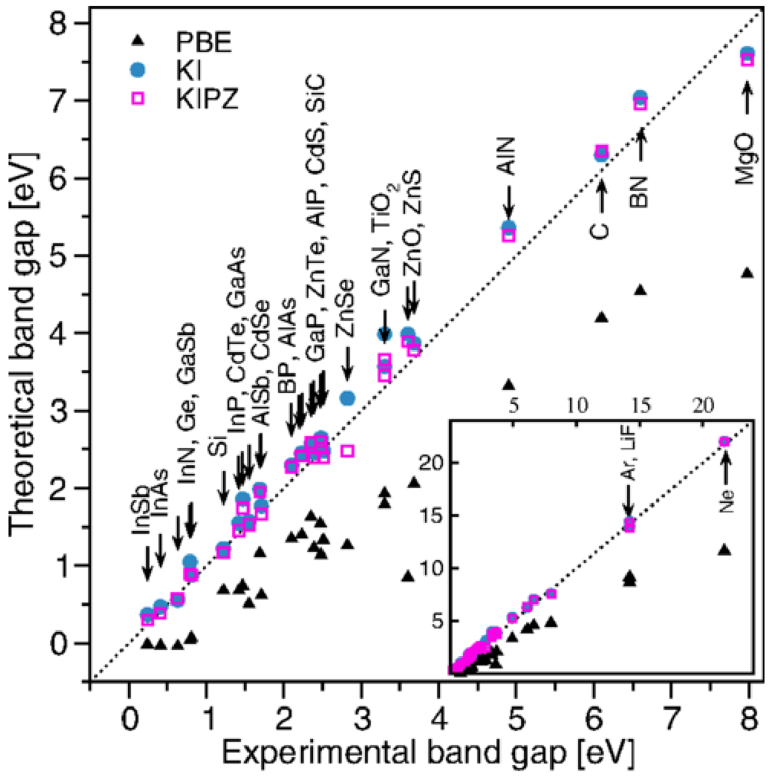


Ultraviolet photoemission spectra



N. Colonna *et al. J. Chem. Theory Comput.* **14,** 2549–2557 (2018), N. L. Nguyen *et al. Phys. Rev. Lett.* **114,** 166405 (2015)

Koopmans functionals: results for solids

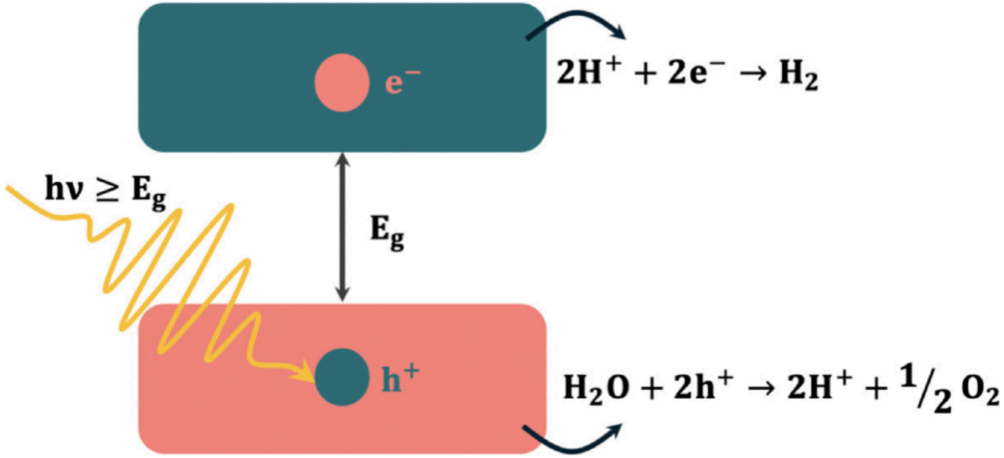


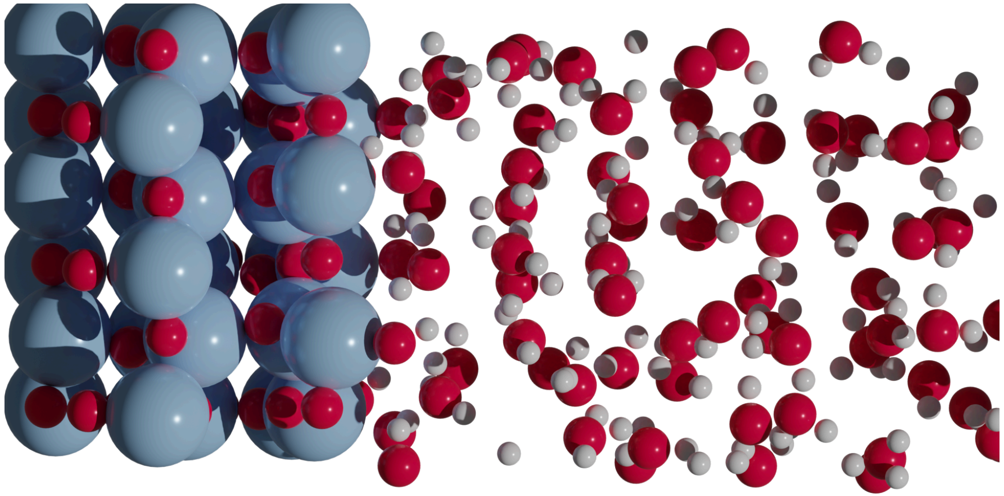
Mean absolute error (eV) across prototypical semiconductors and insulators

	PBE	G_0W_0	**KI**	**KIPZ**	$QSG\widetilde{W}$
E_{gap}	2.54	0.56	**0.27**	**0.22**	0.18
IP	1.09	0.39	**0.19**	**0.21**	0.49

N. L. Nguyen *et al.* *Phys. Rev. X* **8,** 21051 (2018)

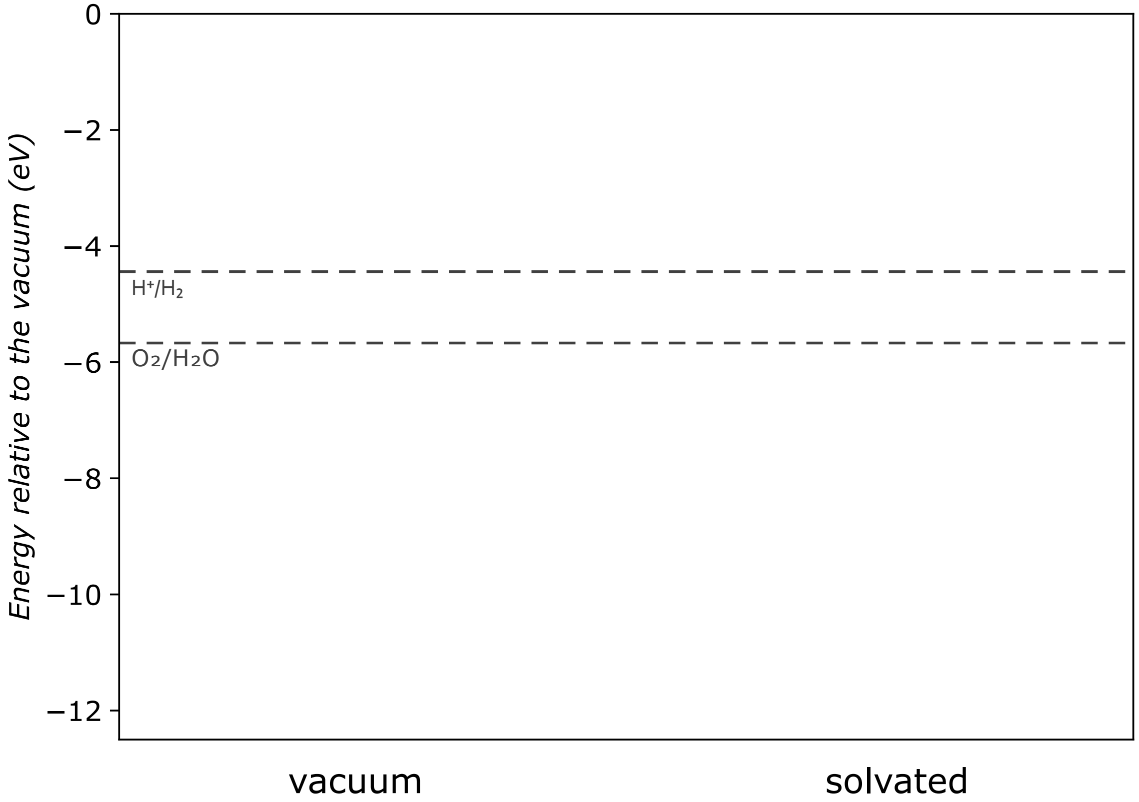
Photocatalytic water-splitting





M. Stojkovic *et al.* *Phys. Rev. Mater.* **10,** 25403 (2026)

Band alignment for water-splitting



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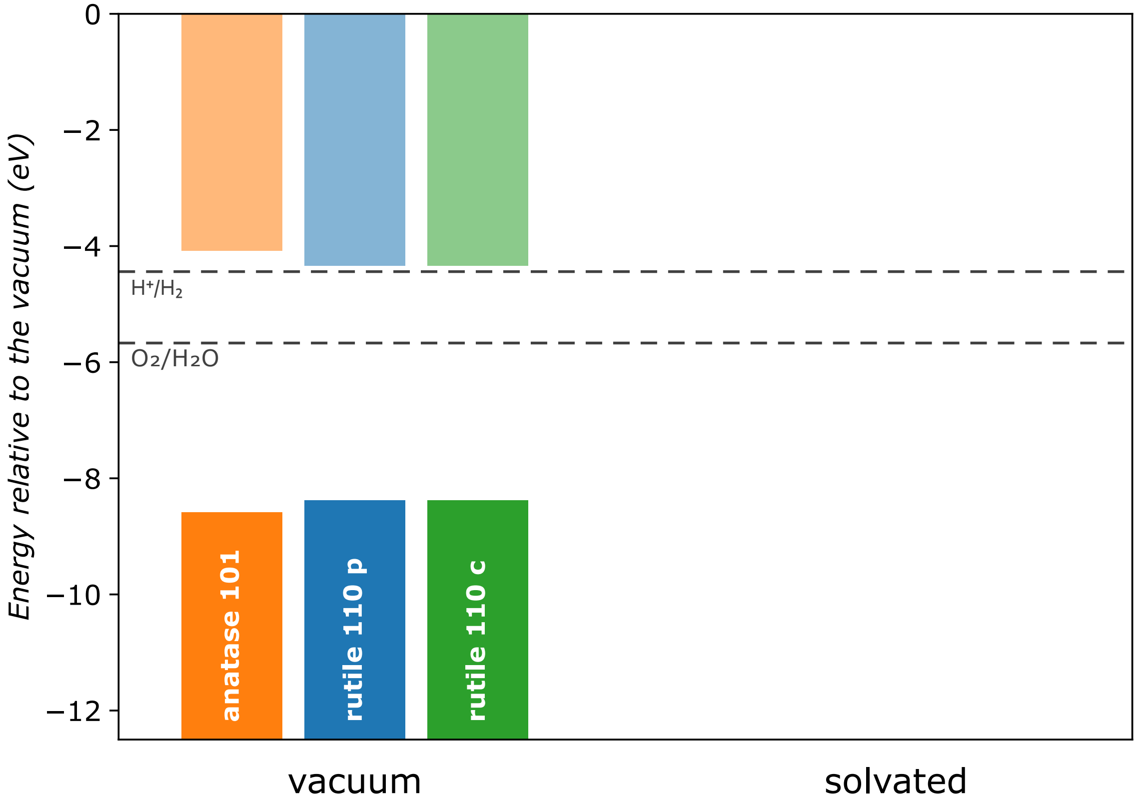

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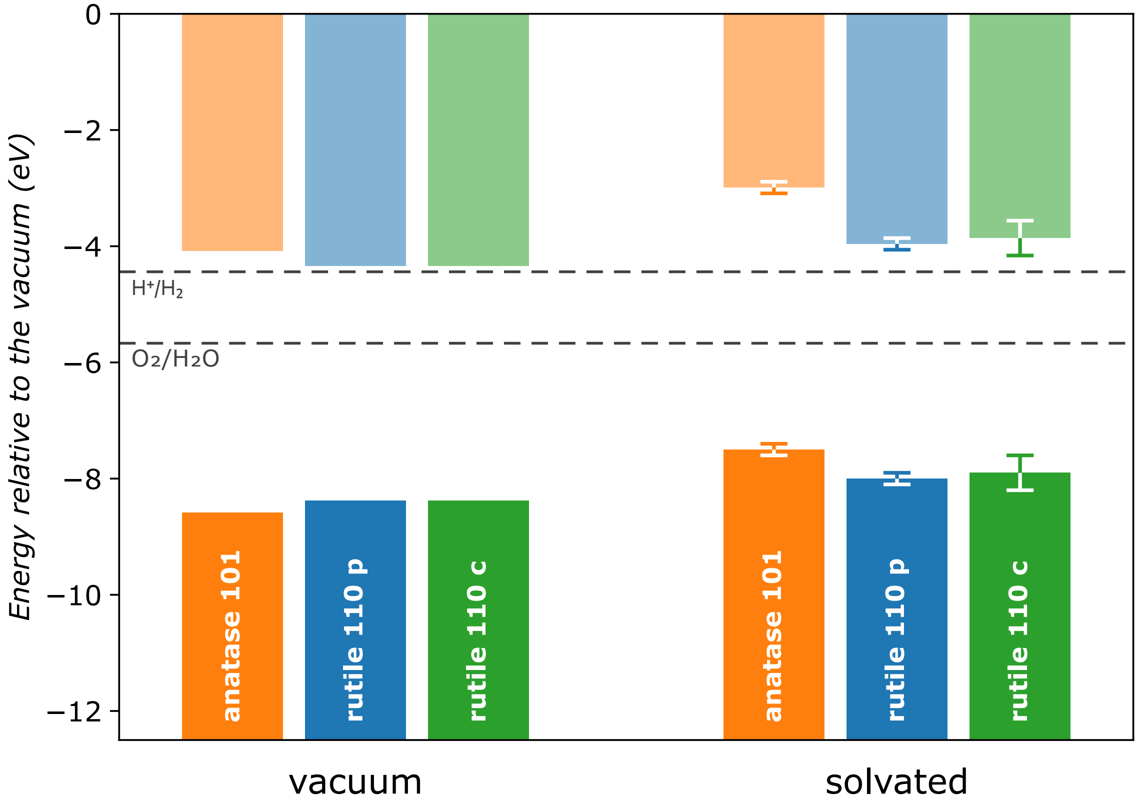

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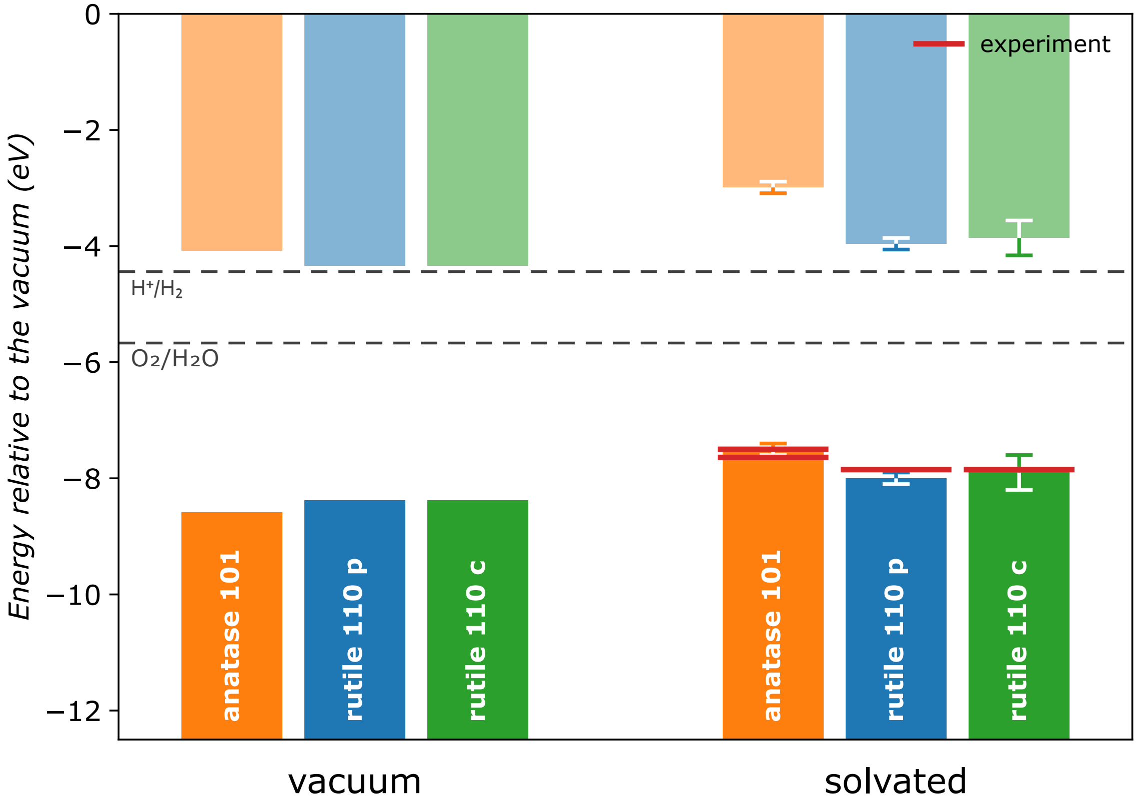
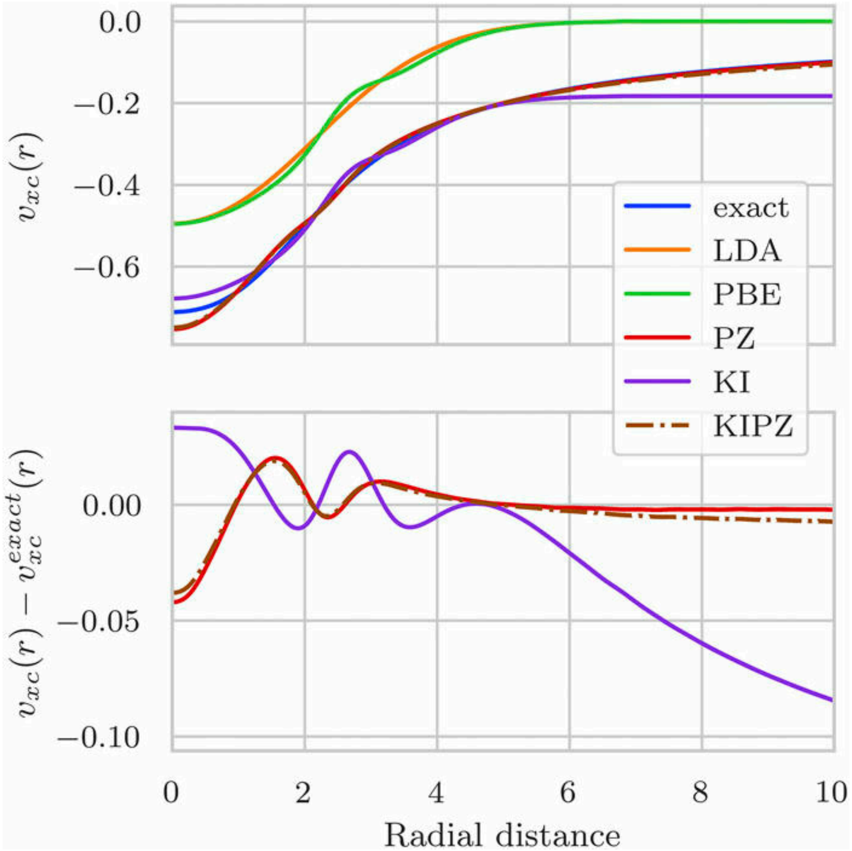


M. Stojkovic *et al. Phys. Rev. Mater.* **10,** 25403 (2026)

Toy systems

For Hooke's atom (two electrons in a harmonic confining potential with Coulombic repulsion)



Y. Schubert *et al. J. Chem. Phys.* **158,** 144113 (2023)

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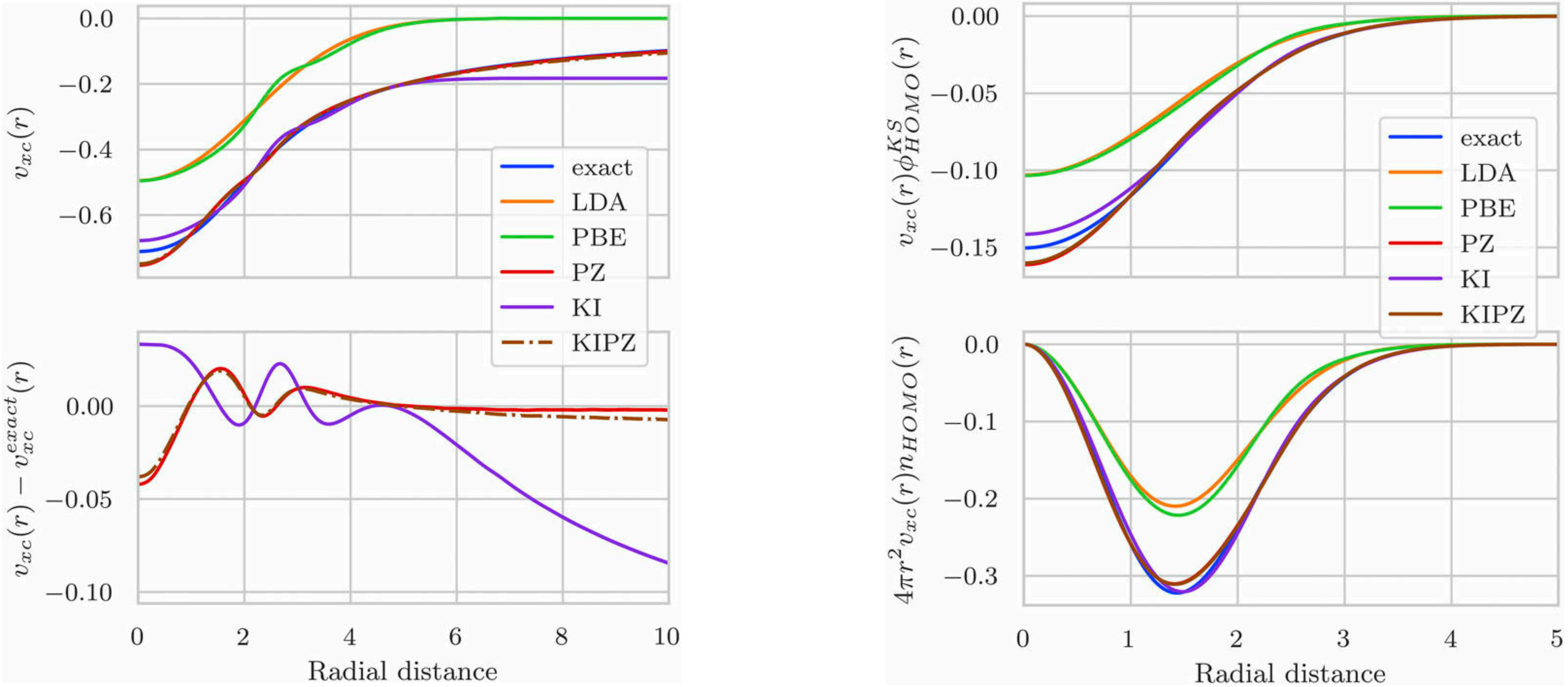

Y. Schubert *et al.* *J. Chem. Phys.* **158,** 144113 (2023)

Koopmans functionals: caveats

N. Q. Su *et al.* *J. Phys. Chem. Lett.* **11,** 1528–1535 (2020)

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Koopmans functionals: caveats

- restricted to systems with a non-zero band gap
- empty state localization in the bulk limit
- can potentially break the crystal point group symmetry

N. Q. Su *et al.* *J. Phys. Chem. Lett.* **11,** 1528–1535 (2020)

Resonance with other efforts

- Wannier transition-state method of Anisimov and Kozhevnikov
- Optimally tuned hybrid functionals of Kronik, Pasquarello, and others (refer back to Leeor's talk on Wednesday)
- Ensemble DFT of Kronik and co-workers
- Koopmans-Wannier of Wang and co-workers
- Dielectric-dependent hybrid functionals of Galli and co-workers
- LOSC functionals of Yang and co-workers

V. I. Anisimov *et al. Phys. Rev. B* **72,** 75125 (2005)

L. Kronik *et al. J. Chem. Theory Comput.* **8,** 1515–1531 (2012), D. Wing *et al. Proc. Natl. Acad. Sci.* **118,** e2104556118 (2021)

E. Kraisler *et al. Phys. Rev. Lett.* **110,** 126403 (2013)

J. Ma *et al. Sci. Rep.* **6,** 24924 (2016)

J. H. Skone *et al. Phys. Rev. B* **93,** 235106 (2016)

C. Li *et al. Natl. Sci. Rev.* **5,** 203–215 (2018)

Computational cost and scaling

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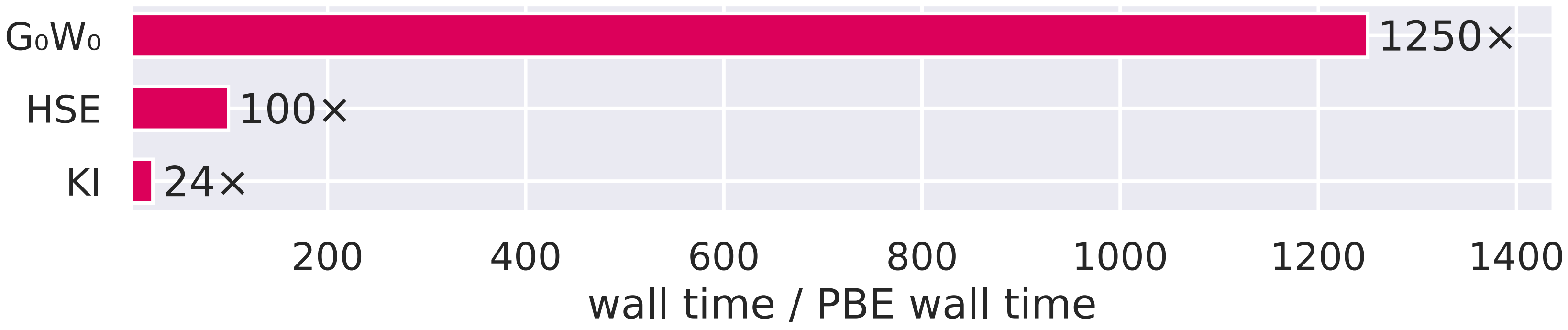

Computational cost and scaling

The vast majority of the computational cost: determining screening parameters

$$\alpha_i = \frac{\langle n_i | \varepsilon^{-1} f_{\mathrm{Hxc}} | n_i \rangle}{\langle n_i | f_{\mathrm{Hxc}} | n_i \rangle}$$

N. L. Nguyen *et al. Phys. Rev. X* **8,** 21051 (2018), R. De Gennaro *et al. Phys. Rev. B* **106,** 35106 (2022)

N. Colonna *et al. J. Chem. Theory Comput.* **18,** 5435–5448 (2022), N. Colonna *et al. J. Chem. Theory Comput.* **14,** 2549–2557 (2018)

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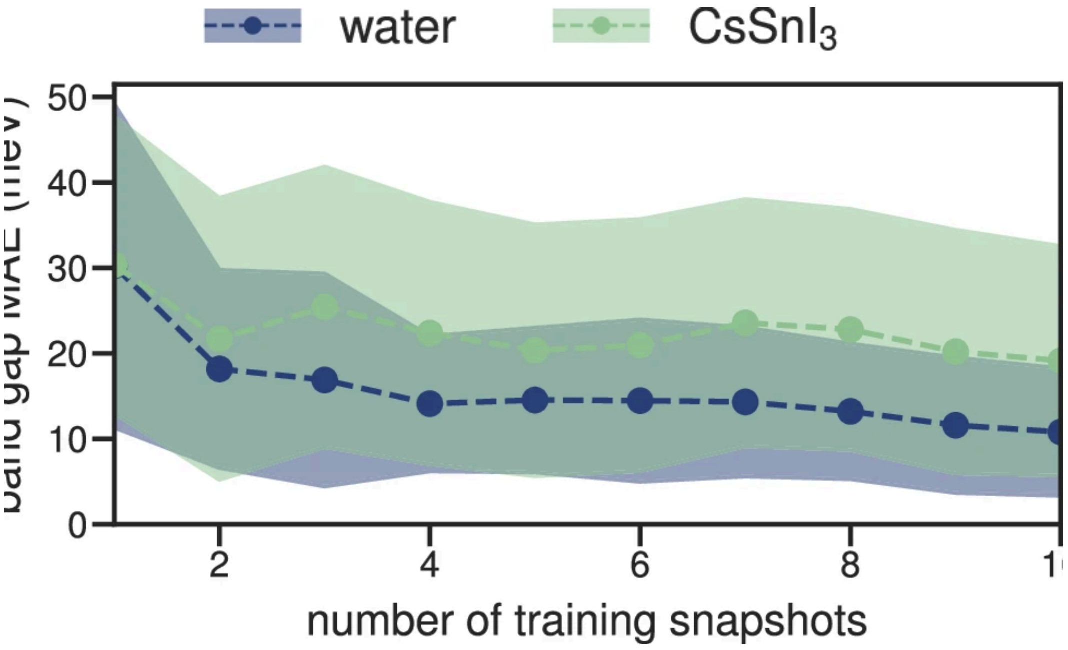
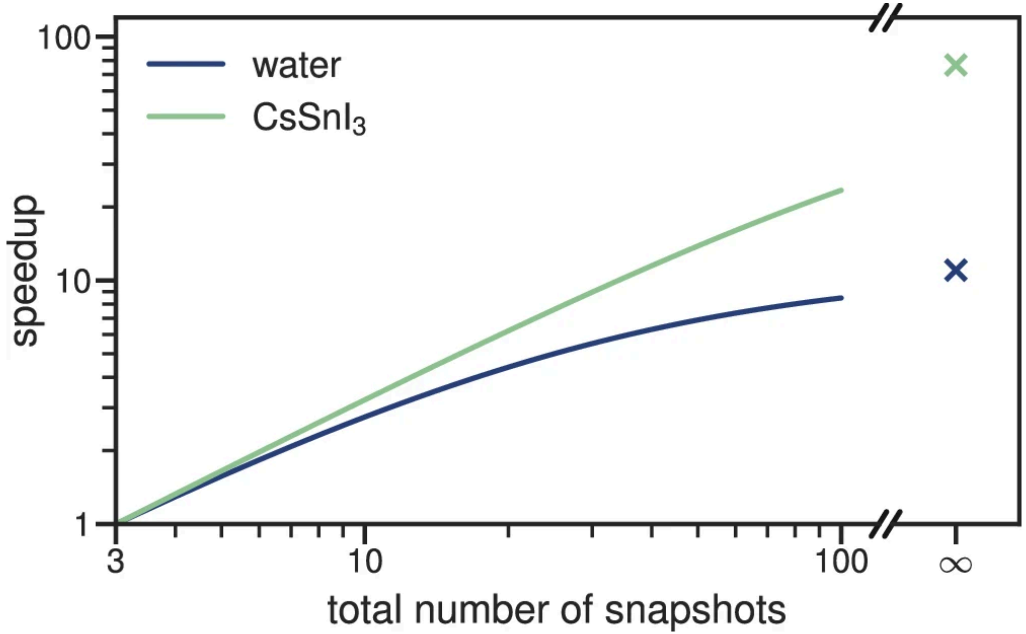
Machine-learned electronic screening



accurate to within $\mathcal{O}$(10 meV) *cf.* typical band gap accuracy of $\mathcal{O}$(100 meV)



speedup of $\mathcal{O}$(10) to $\mathcal{O}$(100)

Y. Schubert *et al. npj Comput Mater* **10,** 1–12 (2024)

Machine-learned electronic screening

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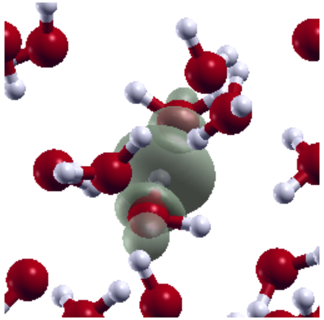

$$\xrightarrow{\text{power spectrum decomposition}} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \end{bmatrix} \xrightarrow{\text{ridge regression}} \alpha_i$$

$$c^i_{nlm,k} = \int \mathrm{d}\boldsymbol{r} g_{nl}(r) Y_{lm}(\theta, \varphi) n^i(\boldsymbol{r} - \boldsymbol{R}^i)$$

$$p^i_{n_1 n_2 l, k_1 k_2} = \pi \sqrt{\frac{8}{2l+1}} \sum_m c^{i*}_{n_1 lm,k_1} c^i_{n_2 lm,k_2}$$

Y. Schubert *et al. npj Comput Mater* **10,** 1–12 (2024)

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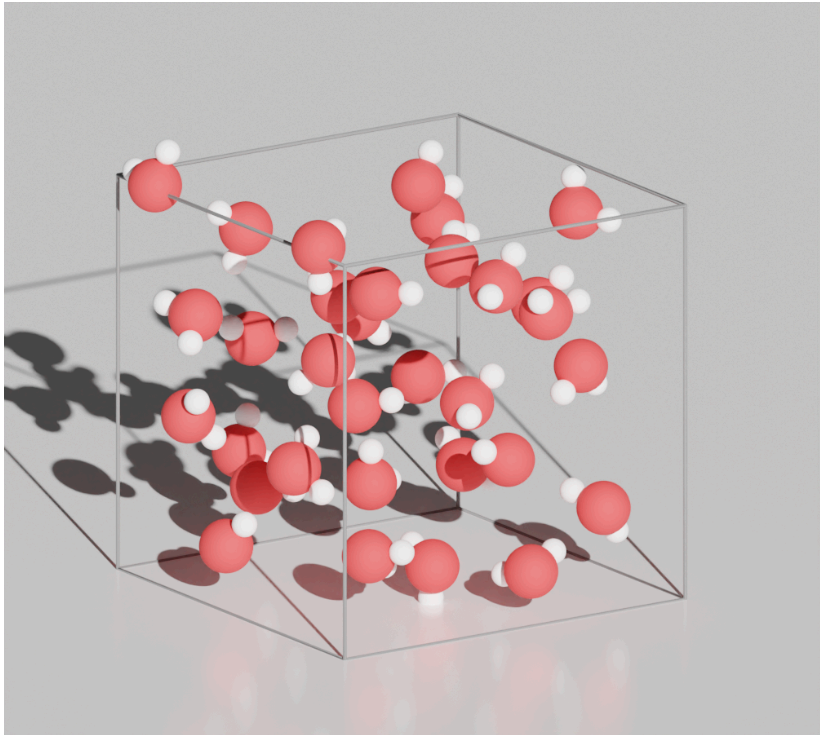
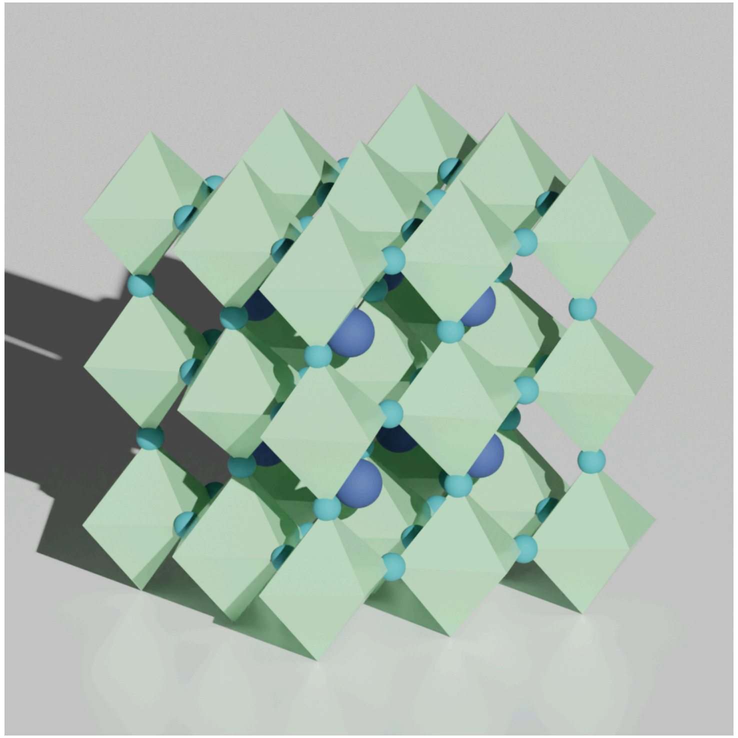


water

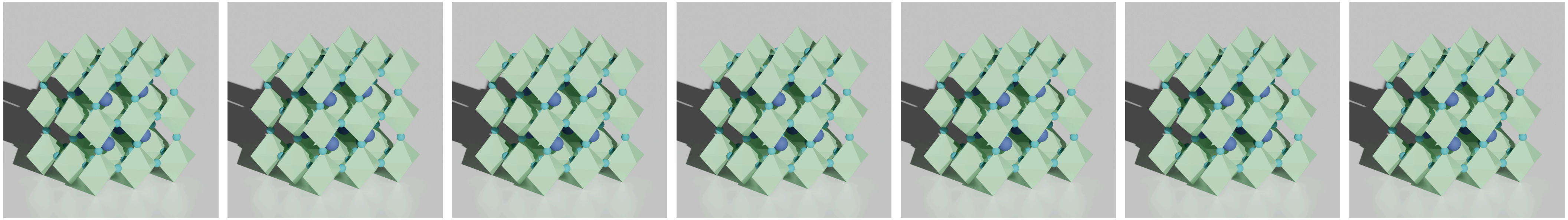


$CsSnI_3$

Y. Schubert *et al.* *npj Comput Mater* **10,** 1–12 (2024)

Machine-learned electronic screening

The use-case

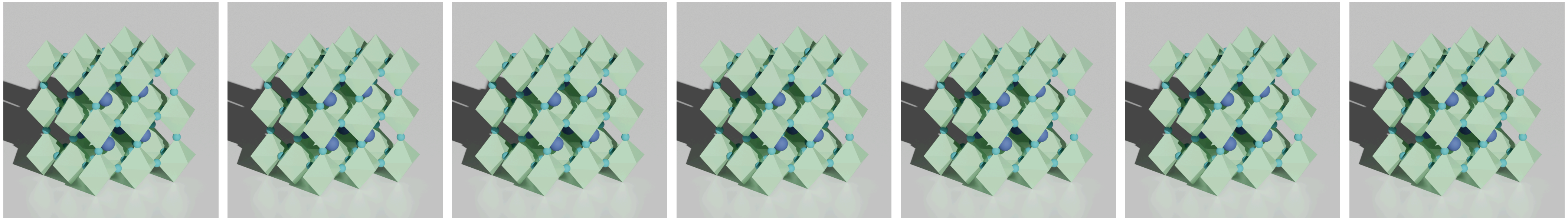


…

train **predict**

Machine-learned electronic screening

The use-case



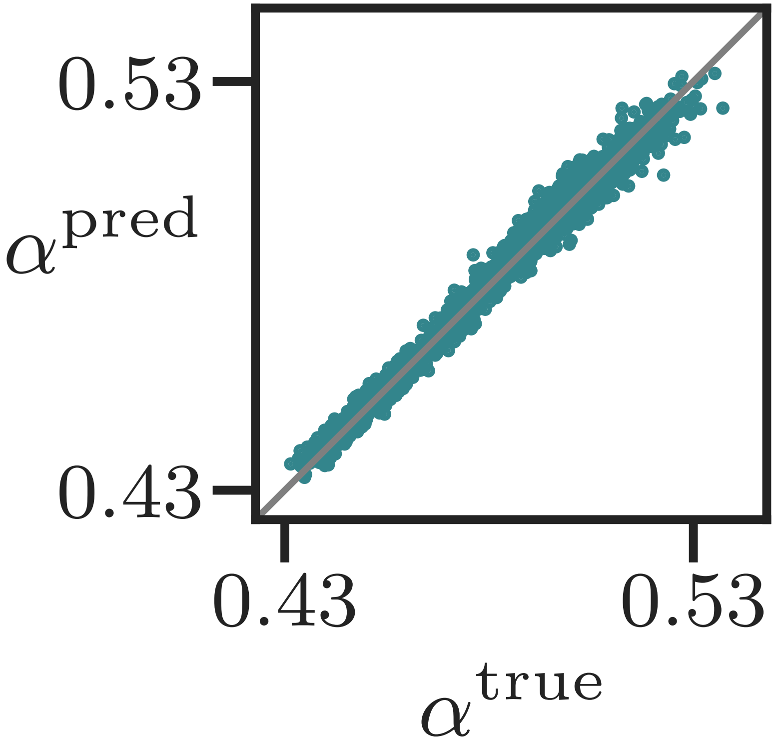
…

train

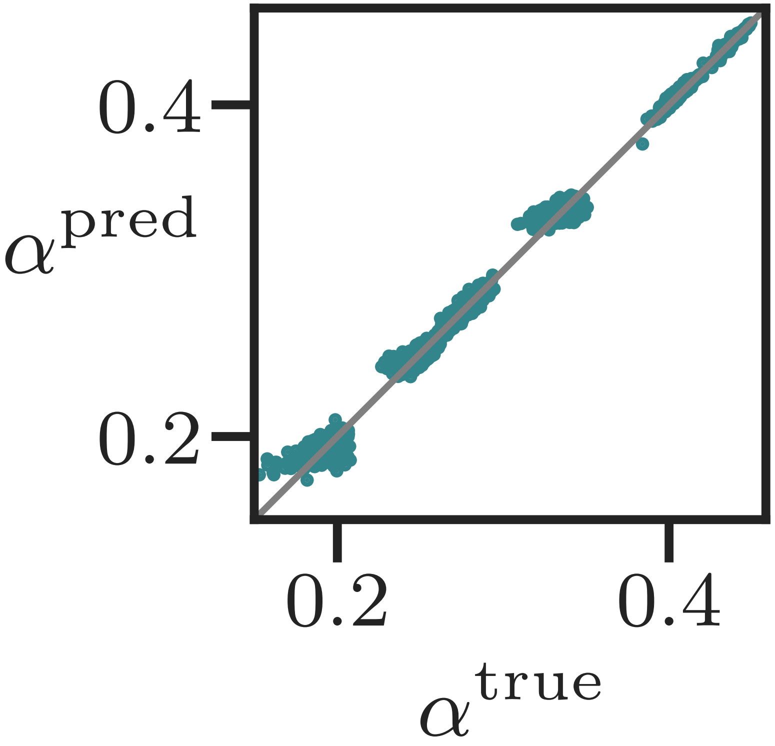
predict

N.B. not a general model

Machine-learned electronic screening



water



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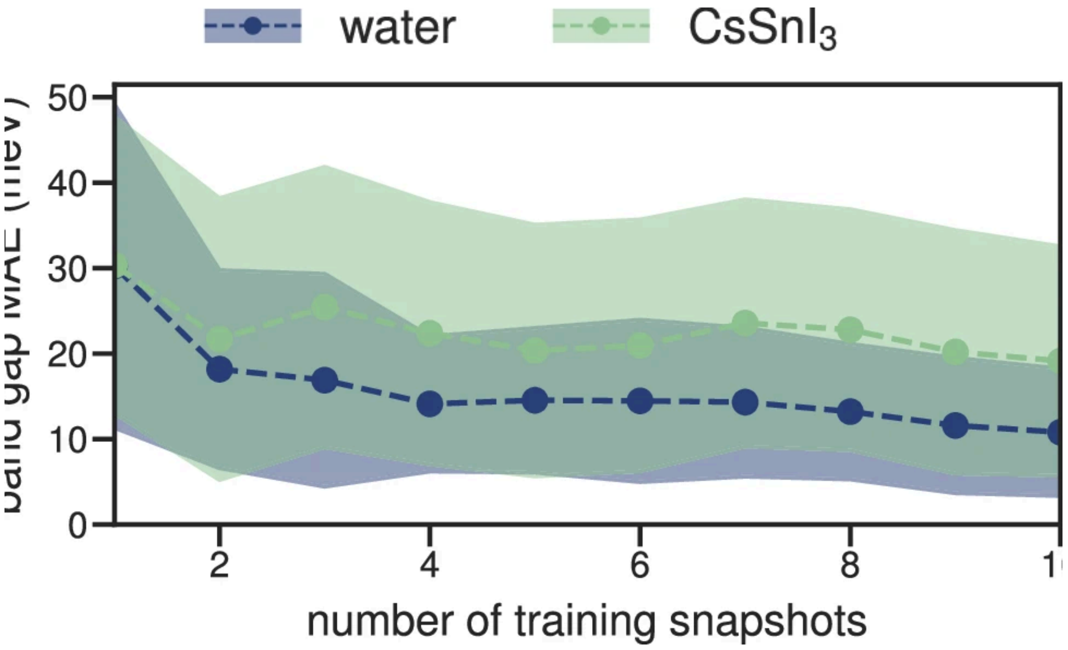

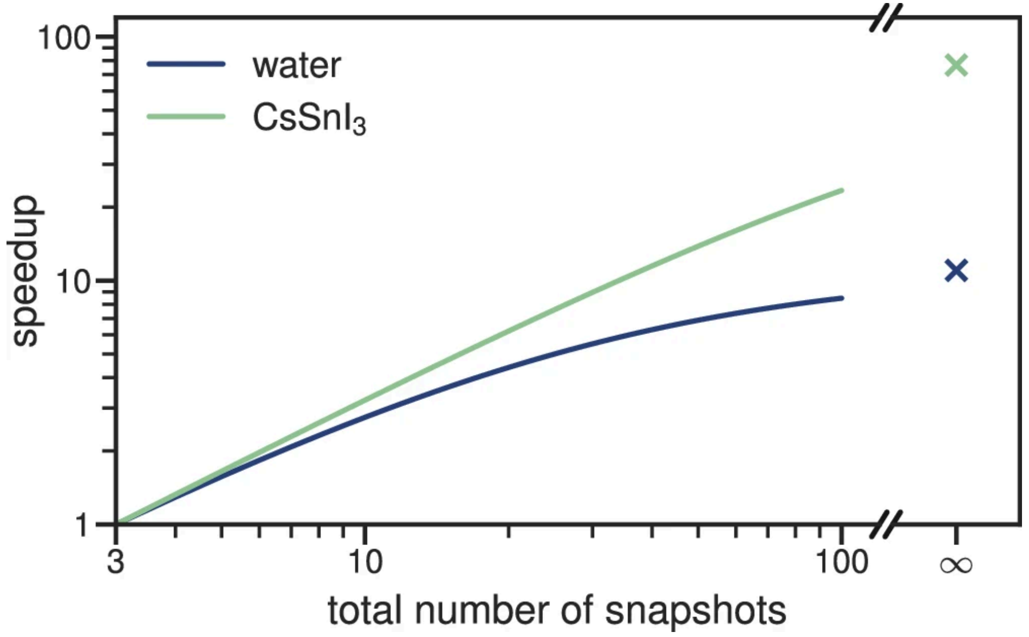

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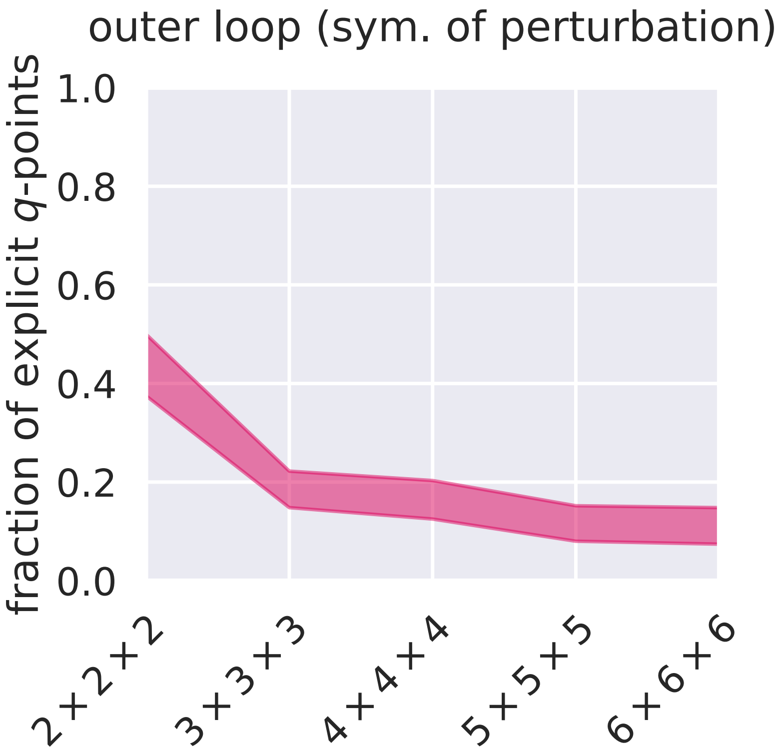
Y. Schubert *et al. npj Comput Mater* **10,** 1–12 (2024)

Taking advantage of symmetries

To compute screening parameters via DFPT…

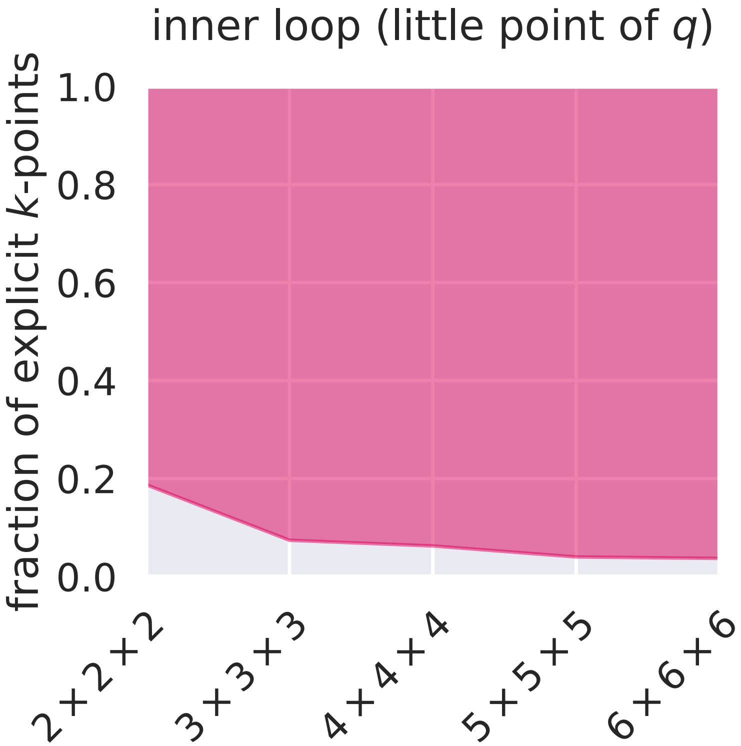
function CalculateAlpha(n)
  for $\boldsymbol{q} \in \mathrm{BZ}$ **do**
    for $\boldsymbol{k} \in \mathrm{BZ}$ **do**
      ▷ Linear system $Ax = b$ to obtain $\Delta\psi_{\boldsymbol{k}+\boldsymbol{q},v}(\boldsymbol{r})$
    end
    $\Delta\rho^{0n}_{\boldsymbol{q}} \leftarrow \sum_{\boldsymbol{k}v} \psi^*_{\boldsymbol{k}v}(\boldsymbol{r})\Delta\psi_{\boldsymbol{k}+\boldsymbol{q},v}(\boldsymbol{r}) + c.c.$
    $\Pi^{(r)}_{0n,\boldsymbol{q}} \leftarrow \langle\Delta\rho^{0n}_{\boldsymbol{q}}|f_{\mathrm{Hxc}}|\rho^{0n}_{\boldsymbol{q}}\rangle$
    $\Pi^{(u)}_{0n,\boldsymbol{q}} \leftarrow \langle\rho^{0n}_{\boldsymbol{q}}|f_{\mathrm{Hxc}}|\rho^{0n}_{\boldsymbol{q}}\rangle$
  end
  return $1 + \sum_{\boldsymbol{q}} \Pi^{(r)}_{0n,\boldsymbol{q}} / \sum_{\boldsymbol{q}} \Pi^{(u)}_{0n,\boldsymbol{q}}$
end

Taking advantage of symmetries



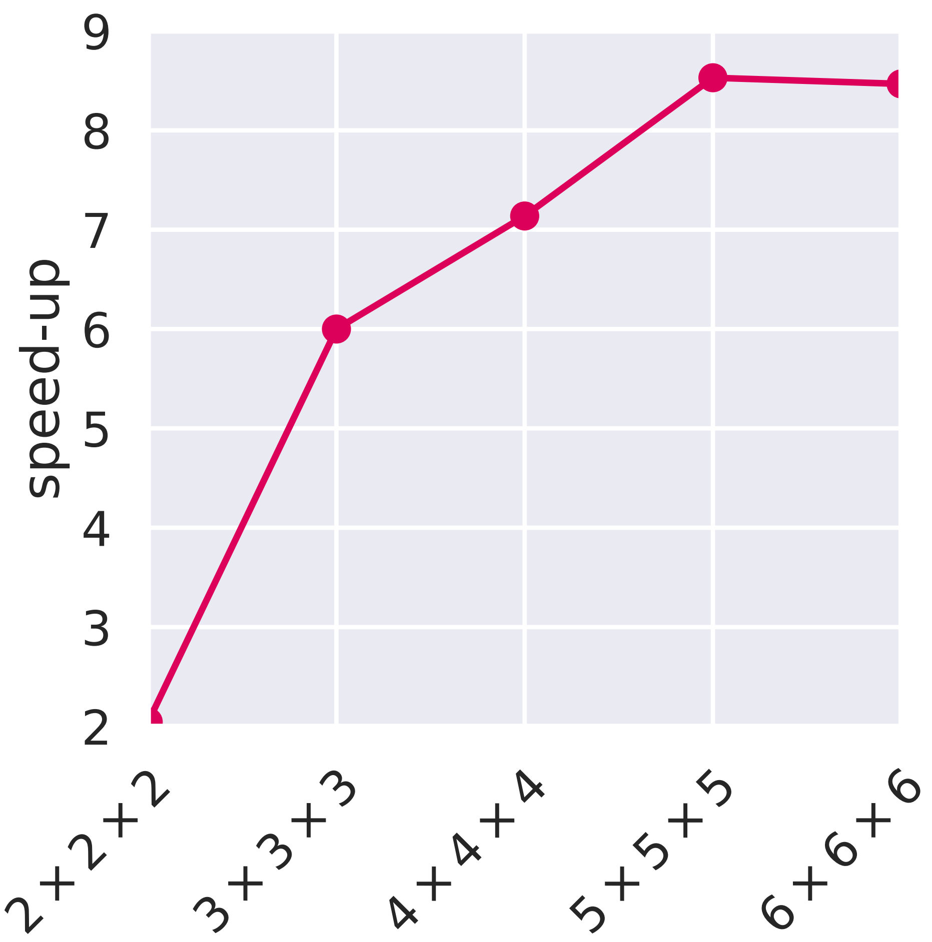
$\boldsymbol{q} \in \mathrm{BZ} \rightarrow \boldsymbol{q} \in \mathrm{IBZ}(n)$ (the symmetry of the perturbation; lower than that of the primitive cell)

Taking advantage of symmetries



$\boldsymbol{k} \in \mathrm{BZ} \rightarrow \boldsymbol{k} \in \mathrm{IBZ}(\boldsymbol{q})$ (can only use symmetries that leave $\boldsymbol{q}$ invariant)

Taking advantage of symmetries



Running a Koopmans functional calculation

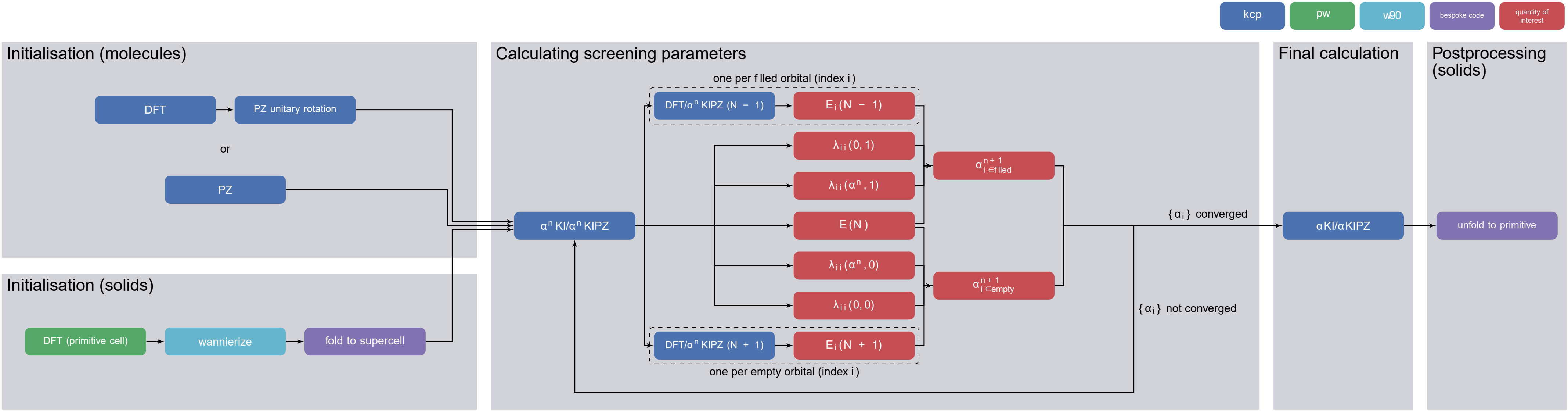
The workflows

The general workflow:

- define/initialize a set of variational orbitals
- calculate the screening parameters $\{\alpha_i\}$
- construct and diagonalize the Hamiltonian

The workflows

(a) finite difference calculations using a supercell



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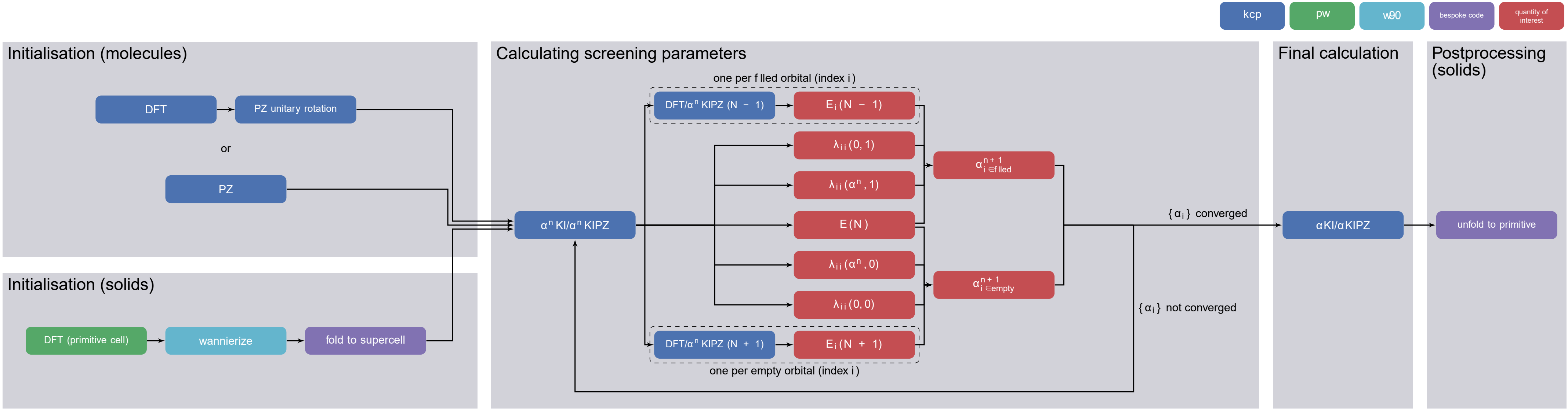
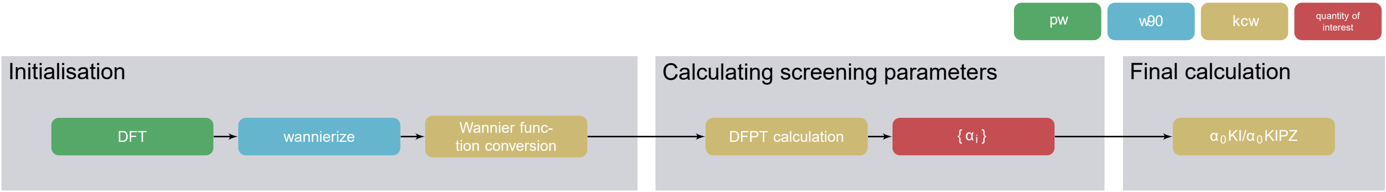


(b) DFPT in a primitive cell



How do I run these calculations?

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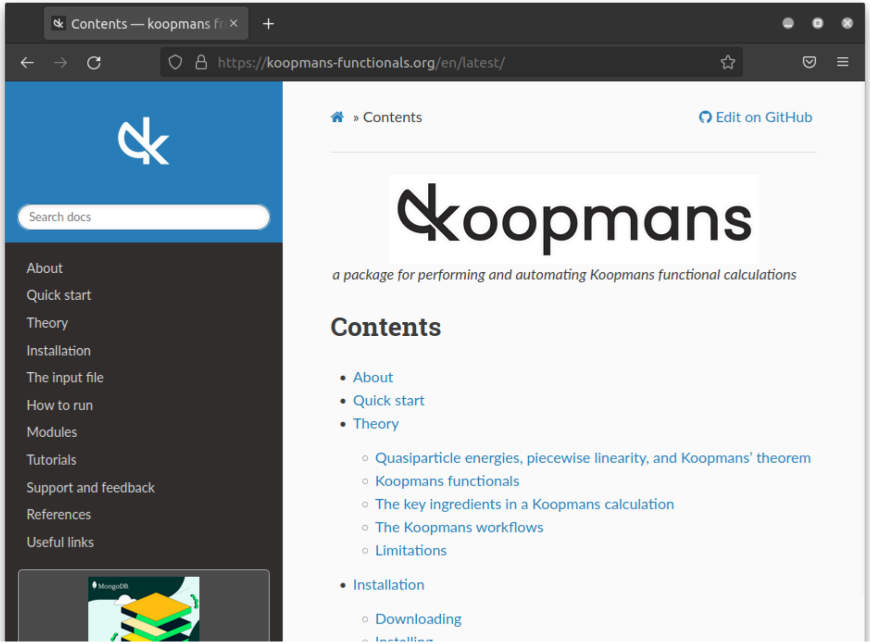
Our solution…

koopmans

The `koopmans` package

- `v1.0` released in 2023 (now `v1.2`)
- implementations of Koopmans functionals
- automated workflows
  - start-to-finish Koopmans calculations
  - Wannierisation
  - dielectric tensor
  - …
- built on top of ASE
- under the hood, calls QUANTUM ESPRESSO
- does not require expert knowledge

`koopmans-functionals.org`



E. B. Linscott *et al.* *J. Chem. Theory Comput.* **19,** 7097–7111 (2023)

A. H. Larsen *et al.* *J. Phys. Condens. Matter* **29,** 273002 (2017)

koopmans: the input file

```json
{
  "workflow": {
    "task": "singlepoint",
    "functional": "ki",
    "method": "dscf",
    "init_orbitals": "mlwfs",
    "alpha_guess": 0.1
  },
  "atoms": {
    "atomic_positions": {
      "units": "crystal",
      "positions": [["Si", 0.00, 0.00, 0.00],
                    ["Si", 0.25, 0.25, 0.25]]
    },
    "cell_parameters": {
      "periodic": true,
      "ibrav": 2,
      "celldm(1)": 10.262
    }
  },
```

```json
  "k_points": {
    "grid": [8, 8, 8],
    "path": "LGXKG"
  },
  "calculator_parameters": {
    "ecutwfc": 60.0,
    "w90": {
      "projections": [
          [{"fsite": [0.125, 0.125, 0.125],
            "ang_mtm": "sp3"}],
          [{"fsite": [0.125, 0.125, 0.125],
            "ang_mtm": "sp3"}]
      ],
      "dis_froz_max": 11.5,
      "dis_win_max": 17.0
    }
  }
}
```

koopmans is scriptable

```python
from ase.build import bulk
from koopmans.kpoints import Kpoints
from koopmans.projections import ProjectionBlocks
from koopmans.workflows import SinglepointWorkflow

# Use ASE to create bulk silicon
atoms = bulk('Si')

# Define the projections for the Wannierization (same for filled and empty manifold)
si_proj = [{'fsite': [0.25, 0.25, 0.25], 'ang_mtm': 'sp3'}]
si_projs = ProjectionBlocks.from_list([si_proj, si_proj], atoms=atoms)

# Create the workflow
workflow = SinglepointWorkflow(atoms = atoms,
        projections = si_projs,
        ecutwfc = 40.0,
        kpoints = Kpoints(grid=[8, 8, 8], path='LGXKG', cell=atoms.cell),
        calculator_parameters = {'pw': {'nbnd': 10},
        'w90': {'dis_froz_max': 10.6, 'dis_win_max': 16.9}})

# Run the workflow
workflow.run()
```

koopmans is scriptable

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but don’t get too used to it…

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# Run the workflow
workflow.run()
```

but don’t get too used to it…

🚧 koopmans v2 is coming… 🚧

- remote compute
- parallel step execution
- provenance-tracking
- error recovery
- and more!

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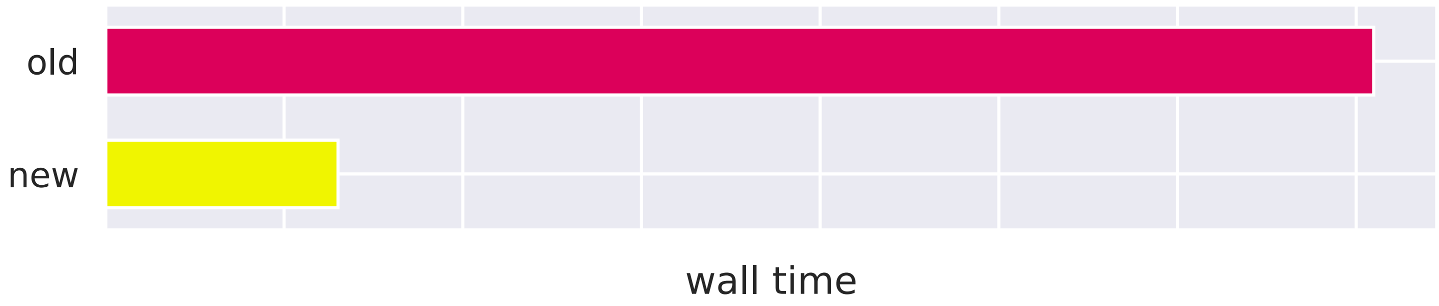
- remote compute
- parallel step execution
- provenance-tracking
- error recovery
- and more!



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Automated Wannierisation

Koopmans functionals rely heavily on Wannier functions…

- to initialise the minmising orbitals, *or*
- in place of the minimising orbitals entirely

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Automated Wannierisation

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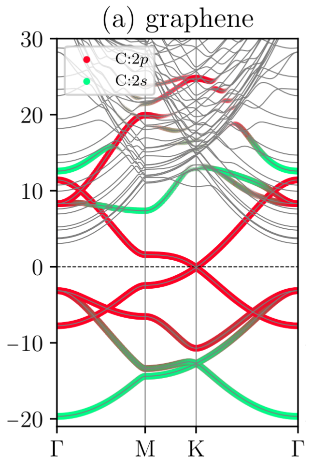
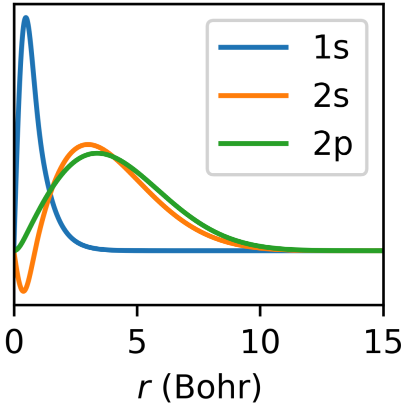
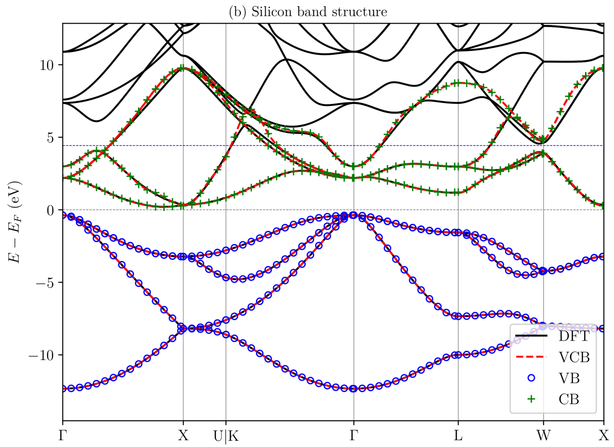


projectability-based disentanglement



use PAOs found in pseudopotentials



parallel transport to separate manifolds

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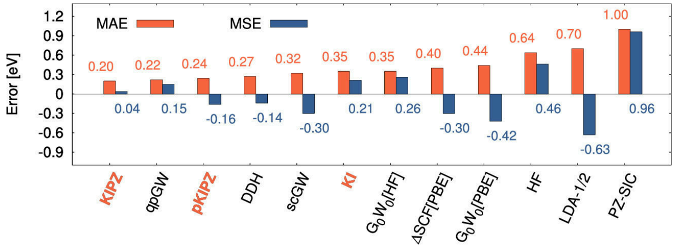
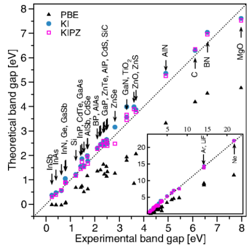
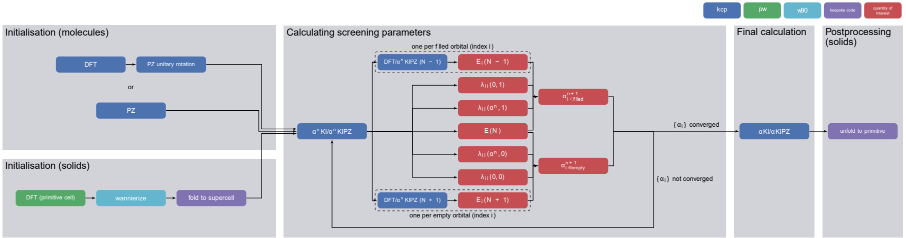
Take home messages







- Koopmans functionals are more complicated than a simple semi-local DFT calculation, because of…
  - orbital-density-dependence
  - screening parameters
- Koopmans functionals are implemented in QUANTUM ESPRESSO
- the complexity of the workflows are handled by the `koopmans` package
- keep your eyes peeled for `v2`!

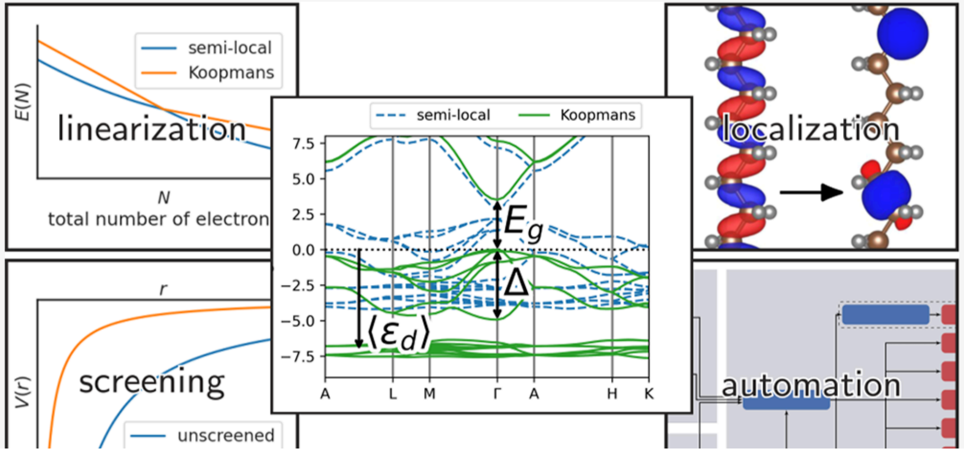
Take home messages

koopmans: An Open-Source Package for Accurately and Efficiently Predicting Spectral Properties with Koopmans Functionals

Edward B. Linscott,*,△ Nicola Colonna,△ Riccardo De Gennaro, Ngoc Linh Nguyen, Giovanni Borghi, Andrea Ferretti, Ismaila Dabo, and Nicola Marzari*

Cite This: https://doi.org/10.1021/acs.jctc.3c00652 Read Online

ABSTRACT: Over the past decade we have developed Koopmans functionals, a computationally efficient approach for predicting spectral properties with an orbital-density-dependent functional framework. These functionals impose a generalized piecewise linearity condition to the entire electronic manifold, ensuring that orbital energies match the corresponding electron removal/addition energy differences (in contrast to semilocal DFT, where a mismatch between the two lies at the heart of the band gap problem and, more generally, the unreliability of Kohn–Sham orbital energies). This strategy has proven to be very powerful,



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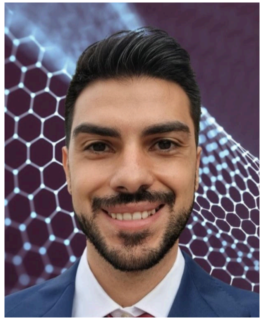

Giovanni Cistaro

Junfeng Qiao

Yannick Schubert

Nicola Marzari

Swiss National Science Foundation

MARVEL

Want to find out more? Go to `koopmans-functionals.org`

slides available at github/elinscott-talks

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Spare slides

Off-diagonal occupancies

PSI

Recap from earlier

Key idea: construct a functional such that the *variational* orbital energies

$$\varepsilon_i^{\text{Koopmans}} = \langle \varphi_i \mid H \mid \varphi_i \rangle = \partial E_{\text{Koopmans}} / \partial f_i$$

are…

- independent of the corresponding occupancies f_i
- equal to the corresponding total energy difference $E_i(N-1) - E(N)$

zero band gap $\rightarrow$ occupancy matrix for variational orbitals is off-diagonal